Supplemental Material

Bayesian inference of stochastic phase dynamics in limit-cycle oscillators

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- EKF
  - linearization assumption must be valid
- Global Phase Filter (no linearization assumption)
  - must be able to "unwrap" phase $\sqrt{2D_\theta \Delta t} \ll 2\pi$
- Phase reduction
  - valid only for ''small noise"
- Reduced model
  - uncorrelated noise
  - noise-free waveform
  - measurement noise vs. amplitude fluctuations

1 Phase Reduction of Noisy Limit-Cycle Oscillators

The reduced stochastic phase model introduced in the main text is motivated by stochastic phase reduction of noisy limit-cycle oscillators.[1–5] The purpose of this section is to relate the effective phase dynamics inferred from noisy observations to the underlying stochastic dynamics in state space.

We consider a dynamical system whose state $x(t) \in \mathbb{R}^d$ evolves according to

$$\dot{x} = f(x) + G\,\xi(t), \tag{1.1}$$

where the deterministic vector field $f(x)$ admits a stable limit cycle $x_0(t)$ of period T_0, and $\xi(t)$ is a vector of independent Gaussian white-noise processes satisfying

$$\langle \xi(t) \rangle = 0, \qquad \langle \xi(t)\xi(t')^\top \rangle = I\,\delta(t-t'). \tag{1.2}$$

The constant matrix G determines the magnitude and correlations of fluctuations in state space. Throughout, we assume that the noise is sufficiently weak that trajectories remain localized near the deterministic limit cycle. The observable signal is

$$y(t) = h(x(t)) + \eta(t), \tag{1.3}$$

where $h(x)$ is an observation function and $\eta(t)$ is Gaussian measurement noise.

As discussed in the main text, the inference framework replaces this state-space description with a reduced stochastic phase model,

$$\dot{\theta} = \omega + \sqrt{2D_\theta}\,\xi_\theta(t). \tag{1.4}$$

Here, the phase variable θ parameterizes position along the oscillation cycle, ω is the effective oscillation frequency, and D_θ is the effective phase diffusivity. The stochastic forcing $\xi_\theta(t)$ is a unit-variance Gaussian white-noise process satisfying

$$\langle \xi_\theta(t) \rangle = 0, \qquad \langle \xi_\theta(t)\xi_\theta(t') \rangle = \delta(t-t'). \tag{1.5}$$

The reduced phase model relates the phase to the observed signal through

$$y(t) \approx y_0(\theta(t)) + \eta(t), \tag{1.6}$$

where $y_0(\theta) = h(x_0(\theta))$ is the deterministic waveform defined on the limit cycle.

The central question is how the effective frequency ω, phase diffusivity D_θ, and inferred phase coordinate θ are related to the underlying stochastic dynamics in state space. We first review classical stochastic phase reduction based on the asymptotic phase and its associated isochrons. We then introduce the projection phase that arises naturally from the observation model used for Bayesian inference and explain why it yields the same long-time phase diffusivity.

1.1 Asymptotic phase reduction

The dynamics of weakly perturbed limit-cycle oscillators can often be reduced to an effective stochastic equation for a single phase variable defined on the deterministic limit cycle. Classical phase reduction theory is based on the asymptotic phase and its associated isochrons, which

foliate the basin of attraction of the limit cycle.[1,2,5] For noisy oscillators, stochastic phase reduction provides effective descriptions of phase diffusion and noise-induced frequency shifts arising from the projection of state-space fluctuations onto the phase direction.[3,4] Here we briefly summarize the weak-noise phase reduction relevant to the generative model introduced above. Specifically, we derive the phase sensitivity function $Z(t)$ and phase-curvature tensor $H(t)$, which determine the effective phase diffusivity D_θ and noise-induced frequency shift $\Delta\omega$, respectively. We then discuss the transverse amplitude dynamics that are neglected in the reduced phase model.

Asymptotic phase and isochrons. Let $x_0(t)$ denote the stable limit cycle of the deterministic system

$$\dot{x} = f(x), \tag{1.7}$$

with period T_0 and angular frequency $\omega_0 = 2\pi/T_0$. In the basin of attraction of the limit cycle, one may define a scalar phase function $\Theta(x)$ that assigns to each state x the phase of the point on the limit cycle approached asymptotically by the trajectory initialized at x.[1,2] On the limit cycle $x_0(t)$, the phase advances uniformly in time,

$$\Theta(x_0(t)) = \omega_0 t + \phi \pmod{2\pi}, \tag{1.8}$$

where ϕ is a constant phase offset. Differentiating with respect to time and applying the chain rule gives

$$\frac{d}{dt}\Theta(x_0(t)) = \nabla\Theta(x_0(t)) \cdot f(x_0(t)) = \omega_0. \tag{1.9}$$

Because the phase evolves uniformly along every deterministic trajectory in the basin of attraction, this relation extends off the limit cycle to give

$$\nabla\Theta(x) \cdot f(x) = \omega_0. \tag{1.10}$$

The level sets of $\Theta(x)$ are known as *isochrons*. Points lying on the same isochron converge asymptotically to the same point on the limit cycle and therefore share the same long-time phase.

Phase sensitivity function, $Z(t)$. The infinitesimal phase response curve (iPRC) is defined as the gradient of the phase function evaluated along the limit cycle,

$$Z(t) := \nabla\Theta(x_0(t)). \tag{1.11}$$

Geometrically, $Z(t)$ quantifies the sensitivity of the oscillator phase to infinitesimal perturbations in state space. Although the phase function $\Theta(x)$ is generally not known analytically, the iPRC can be computed from the defining identity (1.10). Differentiating (1.10) with respect to x_i gives

$$\sum_j \partial_{x_i x_j}\Theta\, f_j + \sum_j \partial_{x_j}\Theta\, \partial_{x_i} f_j = 0. \tag{1.12}$$

Evaluating this expression along the limit cycle and using $\dot{x}_0(t) = f(x_0(t))$ yields

$$\frac{d}{dt}Z_i(t) + \sum_j J_{ji}(t)\, Z_j(t) = 0, \tag{1.13}$$

where $Z_i = \partial_{x_i}\Theta$ and $J_{ji}(t) = \partial_{x_i} f_j(x_0(t))$. In vector form, this adjoint equation becomes

$$\dot{Z}(t) = -J(t)^\top Z(t), \tag{1.14}$$

where $J(t)$ is the Jacobian matrix of the vector field evaluated along the limit cycle. Together with periodicity, $Z(t+T_0) = Z(t)$, and the normalization condition

$$Z(t) \cdot f(x_0(t)) = \omega_0, \tag{1.15}$$

equation (1.14) uniquely determines $Z(t)$. In practice, the stable periodic solution of (1.14) is obtained by integrating the adjoint equation backward in time over many periods and enforcing the normalization condition.

Phase curvature tensor, $H(t)$. To characterize noise-induced corrections to the phase dynamics, we require the second derivatives of the phase function evaluated along the limit cycle,

$$H_{ij}(t) := \partial_{x_i x_j}\Theta(x_0(t)). \tag{1.16}$$

The Hessian matrix $H(t)$ characterizes the local curvature of the phase function, or equivalently the geometry of the isochrons, and determines how fluctuations transverse to the limit cycle generate effective phase drift. As in the first-order case, $H(t)$ is obtained indirectly from the defining identity (1.10). Differentiating (1.10) twice with respect to x_i and x_j gives

$$\sum_k \partial_{x_i x_j x_k}\Theta\, f_k + \sum_k \partial_{x_j x_k}\Theta\, \partial_{x_i} f_k + \sum_k \partial_{x_i x_k}\Theta\, \partial_{x_j} f_k + \sum_k \partial_{x_k}\Theta\, \partial_{x_i x_j} f_k = 0. \tag{1.17}$$

Evaluating this expression along the limit cycle yields

$$\frac{d}{dt}H_{ij}(t) + \sum_k H_{jk}(t)J_{ki}(t) + \sum_k H_{ik}(t)J_{kj}(t) + Q_{ij}(t) = 0, \tag{1.18}$$

where

$$J_{ki}(t) = \partial_{x_i} f_k(x_0(t)), \quad \text{and} \quad Q_{ij}(t) = \sum_k Z_k(t)\, \partial_{x_i x_j} f_k(x_0(t)). \tag{1.19}$$

In matrix form, this becomes

$$\dot{H}(t) = -J(t)^\top H(t) - H(t)J(t) - Q(t) \tag{1.20}$$

Unlike the first-order case, this equation does not uniquely determine $H(t)$. If $H(t)$ is a solution, then so is $H(t) + \alpha\, Z(t)Z(t)^\top$ for any constant α. To obtain a unique solution, we impose periodicity, $H(t+T_0) = H(t)$, together with a gauge condition, for example

$$\frac{1}{T_0}\int_0^{T_0} Z(t)^\top H(t) Z(t)\, dt = 0. \tag{1.21}$$

In practice, $H(t)$ is obtained by integrating the above equation backward in time over one period of the limit cycle, using the previously computed $Z(t)$ and enforcing the gauge condition.

Effective phase diffusivity, D_θ. We now derive the effective dynamics of the phase $\theta(t) = \Theta(x(t))$ for the noisy system. For the deterministic dynamics, the phase evolves according to equation (1.10). In the presence of noise, the state $x(t)$ undergoes small fluctuations about the limit cycle, which induce fluctuations in the phase.

Over a short time interval dt, equation (4.1) implies that the state changes by the increment

$$\delta x = x(t + dt) - x(t) = f(x)\, dt + G\, \delta W, \tag{1.22}$$

where δW is a Gaussian increment with zero mean and covariance

$$\langle \delta W\, \delta W^\top \rangle = I\, dt. \tag{1.23}$$

It follows that the covariance of the state increment is

$$\mathrm{Cov}(\delta x) = C\, dt, \qquad C = GG^\top, \tag{1.24}$$

where C is the noise covariance matrix in state space.

To first order, the corresponding phase increment is obtained by linearizing the phase function,

$$\delta\theta = \Theta(x + \delta x) - \Theta(x) \approx \nabla\Theta(x) \cdot \delta x. \tag{1.25}$$

Geometrically, $\nabla\Theta(x)$ is normal to the isochrons and points in the direction of increasing phase, so that the phase shift is given by the projection of the state increment δx onto this direction. Evaluating the phase gradient along the limit cycle and using $Z(t) = \nabla\Theta(x_0(t))$ gives

$$\delta\theta \approx Z(t) \cdot \delta x, \tag{1.26}$$

where $Z(t)$ defines the local phase direction along the cycle. The deterministic contribution $Z(t) \cdot f(x_0(t))\, dt = \omega_0\, dt$ corresponds to uniform motion along the limit cycle, while the stochastic contribution reflects random displacements in state space projected onto the phase direction. This produces a variance

$$\langle (\delta\theta)^2 \rangle = Z(t)^\top C Z(t)\, dt, \tag{1.27}$$

so that only the component of the noise aligned with $Z(t)$ contributes to phase fluctuations. Averaging over one period then gives the effective phase diffusivity

$$D_\theta = \frac{1}{2T_0} \int_0^{T_0} Z(t)^\top C Z(t)\, dt. \tag{1.28}$$

Noise-induced frequency shift, $\Delta\omega$. At next order, curvature of the phase function leads to a systematic shift in the mean phase velocity. Expanding $\Theta(x)$ to second order,

$$\Theta(x + \delta x) \approx \Theta(x) + Z(t) \cdot \delta x + \frac{1}{2}\, \delta x^\top H(t)\, \delta x, \tag{1.29}$$

shows that, in addition to the linear projection onto the phase direction, quadratic variations in δx produce a nonzero average phase shift. Geometrically, this effect arises from the curvature of the isochrons: fluctuations transverse to the limit cycle are mapped asymmetrically onto phase,

resulting in a systematic drift. Taking the expectation over the noise, and using $\langle \delta x\, \delta x^\top \rangle = C\, dt$, the quadratic term produces a mean increment

$$\langle \delta\theta \rangle = \omega_0\, dt + \frac{1}{2} \operatorname{Tr}(CH(t))\; dt. \tag{1.30}$$

Averaging over one period yields the noise-induced frequency shift

$$\Delta\omega = \frac{1}{2T_0} \int_0^{T_0} \operatorname{Tr}(CH(t))\; dt. \tag{1.31}$$

The above expression corresponds to the weak-noise phase reduction obtained after elimination of fast transverse fluctuations and is consistent with the stochastic phase reduction framework of Yoshimura and Arai.[3]

Amplitude dynamics. The phase reduction relies on a separation between motion along the limit cycle and deviations transverse to it. Let $x_0(t)$ denote the limit cycle, with tangent vector $\dot{x}_0(t) = f(x_0(t))$. The phase direction is defined by the adjoint vector $Z(t)$, which satisfies $Z(t) \cdot \dot{x}_0(t) = \omega_0$ and projects state perturbations onto phase. We decompose the dynamics near the limit cycle as

$$x(t) = x_0(\theta(t)) + u(t), \tag{1.32}$$

where $u(t)$ represents deviations from the cycle satisfying $Z(\theta(t)) \cdot u(t) = 0$, so that u lies in the $(d-1)$-dimensional subspace transverse to the phase direction. Starting from the Langevin equation (4.1) and linearizing about the limit cycle yields

$$\dot{u} = J(t)\, u + P(t)\, G\, \xi(t), \tag{1.33}$$

where $J(t) = Df(x_0(t))$ and

$$P(t) = I - \frac{f(x_0(t))\, Z(t)^\top}{\omega_0} \tag{1.34}$$

is the projection operator onto the transverse subspace (satisfying $Z(t)^\top P(t) = 0$). Thus, only the component of the noise orthogonal to the phase direction drives amplitude fluctuations.

By stability of the limit cycle, small perturbations transverse to the cycle decay exponentially under the linearized dynamics. The linear system (1.33) has one neutral direction associated with motion along the cycle, while all remaining directions decay with rates determined by the Floquet exponents. The disturbance $u(t)$ obeys a linear stochastic system with stable drift and additive forcing. This is the time-periodic analogue of an Ornstein–Uhlenbeck process, in which fluctuations are continuously driven by noise but remain bounded due to linear relaxation. As a result, $u(t)$ has a stationary distribution with typical magnitude set by a balance between the noise amplitude G and the transverse relaxation rates, and a correlation time set by the inverse decay rates of the transverse modes.

Limits of validity.

1.2 Projection-based phase coordinates

While asymptotic phase reduction provides a natural dynamical description of weakly perturbed oscillators, it is not always the most convenient representation for inference from noisy observations. In many experimental settings, the observed oscillation is more naturally described relative to the deterministic waveform itself. This motivates an alternative phase coordinate based on projection onto the deterministic limit cycle. Unlike the asymptotic phase, which is defined through long-time convergence to isochrons, the projection phase is defined geometrically from the instantaneous state of the oscillator.

Projection phase, θ_{p}, on the deterministic limit cycle. The reduced observation model introduced above represents the measured signal using the deterministic waveform

$$y_0(\theta) := h(x_0(\theta)), \tag{1.35}$$

obtained by evaluating the measurement function along the deterministic limit cycle. This approximation assumes that fluctuations transverse to the limit cycle produce only small perturbations to the observed waveform. Using the decomposition in equation (1.32),

$$y(t) = h(x_0(\theta) + u) + \eta(t) = y_0(\theta(t)) + \eta(t) + \mathcal{O}(u) \tag{1.36}$$

where the neglected terms describe stochastic waveform fluctuations and noise-induced distortions arising from the transverse dynamics. Under this approximation, the observations are represented by the reduced model

$$y_k \approx y_0(\theta_k^{\mathrm{p}}) + \eta_k, \tag{1.37}$$

where the residual term η_k now incorporates both measurement noise and the neglected effects of transverse fluctuations. Assuming these residuals are approximately Gaussian with covariance R, the projection phase is defined by the maximum-likelihood estimate

$$\theta_{\mathrm{p}} = \arg\min_{\phi} \big(y - y_0(\phi)\big)^{\top} R^{-1} \big(y - y_0(\phi)\big). \tag{1.38}$$

Thus, θ_{p} corresponds to a projection onto the deterministic waveform using the metric induced by the covariance matrix R.

Dynamics of the projection phase. Unlike the asymptotic phase, the projection phase does not eliminate transverse fluctuations dynamically. Instead, deviations from the deterministic limit cycle contribute directly to the inferred phase through the projection geometry in observation space. As a result, the projection phase inherits temporal correlations associated with the transverse relaxation dynamics.

By construction, the asymptotic phase θ and projection phase θ_{p} assign the same phase to points on the deterministic limit cycle. Their differences arise only for trajectories displaced away from the cycle by transverse fluctuations. Away from the cycle, transverse fluctuations modify the projected phase through the local geometry of the waveform. Writing the stochastic dynamics as

$$x(t) = x_0(\theta(t)) + u(t), \tag{1.39}$$

the projection phase may be expanded locally as

$$\theta_{\mathrm{p}}(t) = \theta(t) + \delta\theta(u(t)), \tag{1.40}$$

where $\delta\theta(u)$ represents the phase shift induced by transverse deviations from the limit cycle. Because the transverse fluctuations $u(t)$ evolve according to the stable stochastic dynamics discussed above, the correction $\delta\theta(u(t))$ exhibits temporal correlations with a characteristic decay time determined by the transverse Floquet relaxation rates.

At short times, the projection phase is dominated by direct geometric projection of the state-space fluctuations onto the tangent direction of the deterministic waveform. For isotropic state-space diffusion with diffusivity D_x, a local tangential displacement satisfies

$$\delta x = \partial_\theta x_0(\theta)\, \delta\theta_{\mathrm{p}}, \tag{1.41}$$

so that projection onto the tangent direction gives

$$\delta\theta_{\mathrm{p}} = \frac{\partial_\theta x_0(\theta) \cdot \delta x}{\partial_\theta x_0(\theta) \cdot \partial_\theta x_0(\theta)}. \tag{1.42}$$

Using

$$\langle \delta x\, \delta x^\top \rangle = 2D_x I\, dt, \tag{1.43}$$

the local short-time diffusivity of the projection phase becomes

$$D_{\theta_{\mathrm{p}}}^{\mathrm{short}}(\theta) = \frac{D_x}{\partial_\theta x_0(\theta) \cdot \partial_\theta x_0(\theta)}. \tag{1.44}$$

Parameterizing the limit cycle by time yields

$$D_{\theta_{\mathrm{p}}}^{\mathrm{short}}(t) = D_x \frac{\omega_0^2}{\dot{x}_0(t) \cdot \dot{x}_0(t)}, \tag{1.45}$$

and averaging over one cycle gives

$$D_{\theta_{\mathrm{p}}}^{\mathrm{short}} = \frac{1}{T_0} \int_0^{T_0} D_x \frac{\omega_0^2}{\dot{x}_0(t) \cdot \dot{x}_0(t)}\, dt. \tag{1.46}$$

At longer times, however, the transverse fluctuations relax toward their stationary distribution and the associated temporal correlations decay. On timescales large compared to the transverse relaxation time, the projection phase loses memory of the instantaneous geometric projection and its dynamics become dominated by the irreversible phase diffusion of the asymptotic phase. Consequently, the long-time growth rate of the drift-subtracted mean squared displacement satisfies

$$\lim_{\tau\to\infty} \frac{\left\langle \left(\theta_{\mathrm{p}}(t+\tau) - \theta_{\mathrm{p}}(t) - \omega\tau\right)^2 \right\rangle}{2\tau} = D_\theta, \tag{1.47}$$

so that the asymptotic and projection phases share the same long-time diffusivity.

Consequently, the projection phase is generally not described by a simple Markovian diffusion process at short times. Instead, its increments contain both diffusive contributions arising from asymptotic phase fluctuations and correlated contributions arising from transient excursions away from the deterministic waveform. The crossover between the short-time diffusivity $D_{\theta_{\mathrm{p}}}^{\mathrm{short}}$ and the asymptotic diffusivity D_θ occurs on the timescale of transverse relaxation to the deterministic limit cycle.

1.3 Example: Stuart–Landau oscillator

As a concrete example, we consider the Stuart–Landau oscillator, the normal form of a supercritical Hopf bifurcation. The deterministic dynamics are given by

$$f(x) = \begin{pmatrix} a(1-r^2)x_1 - [1+b(r^2-1)]x_2 \\ a(1-r^2)x_2 + [1+b(r^2-1)]x_1 \end{pmatrix}, \tag{1.48}$$

where $x = (x_1, x_2)^\top$ and $r^2 = x_1^2 + x_2^2$. In polar coordinates (r, φ), the dynamics decouple as

$$\dot{r} = a(r - r^3), \tag{1.49}$$

$$\dot{\varphi} = 1 + b(r^2 - 1), \tag{1.50}$$

so that the deterministic system admits a stable limit cycle at $r = 1$ with angular frequency $\omega_0 = 1$.

The parameter $a > 0$ controls the rate of radial relaxation toward the limit cycle, while b determines the amplitude dependence of the angular velocity. Consequently, the ratio b/a quantifies the extent to which transient amplitude fluctuations are converted into phase shifts before relaxing back to the cycle.

In this example, we consider the simplest observation model,

$$h(x) = x, \tag{1.51}$$

so that the full state is observed directly up to additive Gaussian measurement noise. This allows the asymptotic and projection phase coordinates to be compared explicitly for the same noisy trajectory.

For the Stuart–Landau oscillator, the projection phase is the polar angle,

$$\theta_{\mathrm{p}} = \varphi, \tag{1.52}$$

whereas the asymptotic phase is

$$\theta = \varphi + \frac{b}{a} \ln r. \tag{1.53}$$

Thus, radial displacements away from the deterministic limit cycle change the asymptotic phase but not the projection phase.

Near the deterministic limit cycle, writing

$$r(t) = 1 + \rho(t), \qquad |\rho| \ll 1, \tag{1.54}$$

gives

$$\theta(t) \approx \theta_{\mathrm{p}}(t) + \frac{b}{a}\rho(t). \tag{1.55}$$

Equivalently,

$$\theta_{\mathrm{p}}(t) \approx \theta(t) - \frac{b}{a}\rho(t). \tag{1.56}$$

The projection phase therefore differs from the asymptotic phase by a temporally correlated amplitude-dependent contribution.

For isotropic state-space diffusion with diffusivity D_x, the asymptotic phase undergoes effective diffusion with diffusivity

$$D_\theta = \left(1 + \frac{b^2}{a^2}\right) D_x, \tag{1.57}$$

where the first term corresponds to direct tangential diffusion along the limit cycle, while the second arises from the conversion of amplitude fluctuations into phase fluctuations through the amplitude–phase coupling term $(b/a)\ln r$. The corresponding noise-induced frequency shift is

$$\Delta\omega = \frac{b}{a} D_x. \tag{1.58}$$

For the Stuart–Landau oscillator, the radial fluctuations obey an Ornstein–Uhlenbeck process near the deterministic limit cycle,

$$\dot{\rho} = -2a\,\rho + \sqrt{2D_x}\,\xi_r(t), \tag{1.59}$$

with relaxation time $(2a)^{-1}$ and stationary variance

$$\langle \rho^2 \rangle = \frac{D_x}{2a}. \tag{1.60}$$

Thus, the parameter a controls both the magnitude and correlation time of amplitude fluctuations.

Because the projection phase differs from the asymptotic phase by the correlated fluctuation term $-(b/a)\rho(t)$, the drift-subtracted mean squared displacement (MSD) of the projection phase exhibits two distinct regimes. At short times,

$$\left\langle \big(\theta_{\mathrm{p}}(t+\tau) - \theta_{\mathrm{p}}(t) - \omega\tau\big)^2 \right\rangle \sim 2D_x\tau, \qquad \tau \ll (2a)^{-1}, \tag{1.61}$$

reflecting direct tangential diffusion along the deterministic limit cycle. At longer times, however, the radial fluctuations relax and the projection phase approaches the same irreversible phase diffusion as the asymptotic phase. Consequently,

$$\left\langle \big(\theta_{\mathrm{p}}(t+\tau) - \theta_{\mathrm{p}}(t) - \omega\tau\big)^2 \right\rangle \sim 2D_\theta\tau, \qquad \tau \gg (2a)^{-1}, \tag{1.62}$$

with

$$D_\theta = \left(1 + \frac{b^2}{a^2}\right) D_x. \tag{1.63}$$

Thus, the projection phase initially captures only the instantaneous geometric projection of noise onto the tangent direction of the deterministic waveform, whereas the asymptotic phase isolates the irreversible long-time phase diffusion. The crossover between these regimes occurs on the transverse relaxation timescale $(2a)^{-1}$.

2 Example 1a: Synthetic phase data with linear measurement

We begin with a minimal synthetic benchmark that illustrates our inference procedure for a stochastic phase oscillator observed through linear measurements. The latent variable is the oscillator phase $\theta(t)$, which evolves according to a biased random walk with constant angular frequency ω and phase diffusivity D_θ. The observed data consist of noisy measurements of this phase.

Given N measurements $y_{1:N} = \{y_1, \ldots, y_N\}$ obtained at uniformly spaced times $t_k = k\Delta t$, we seek to infer the angular frequency ω, phase diffusivity D_θ, and the measurement noise variance R. In addition to these model parameters $\lambda = \{\omega, D_\theta, R\}$, we infer the latent phase trajectory $\theta_{1:N} = \{\theta_1, \ldots, \theta_N\}$. Throughout, we adopt a Bayesian perspective in which both model parameters and latent variables are treated probabilistically.

2.1 Generative Model

The latent phase evolves according to the discrete-time stochastic process

$$\theta_{k+1} = \theta_k + \omega\Delta t + \varepsilon_k, \quad \varepsilon_k \sim \mathcal{N}(0, 2D_\theta\Delta t), \quad k = 1, \ldots, N-1 \tag{2.1}$$

The measurement model is linear

$$y_k = \theta_k + \eta_k, \quad \eta_k \sim \mathcal{N}(0, R), \quad k = 1, \ldots, N \tag{2.2}$$

The joint distribution for the latent phases and the noisy observations—also known as the complete-data likelihood—is therefore written as

$$p(\theta_{1:N}, y_{1:N} \mid \omega, D_\theta, R) = p(\theta_1)\left[\prod_{k=2}^{N} p(\theta_k \mid \theta_{k-1}, \omega, D_\theta)\right]\left[\prod_{k=1}^{N} p(y_k \mid \theta_k, R)\right] \tag{2.3}$$

The structure of this model is illustrated by the directed graph shown in Fig. S1a. There are two types of arrows that denote different conditional distributions: (1) the **process model**, $p(\theta_k \mid$

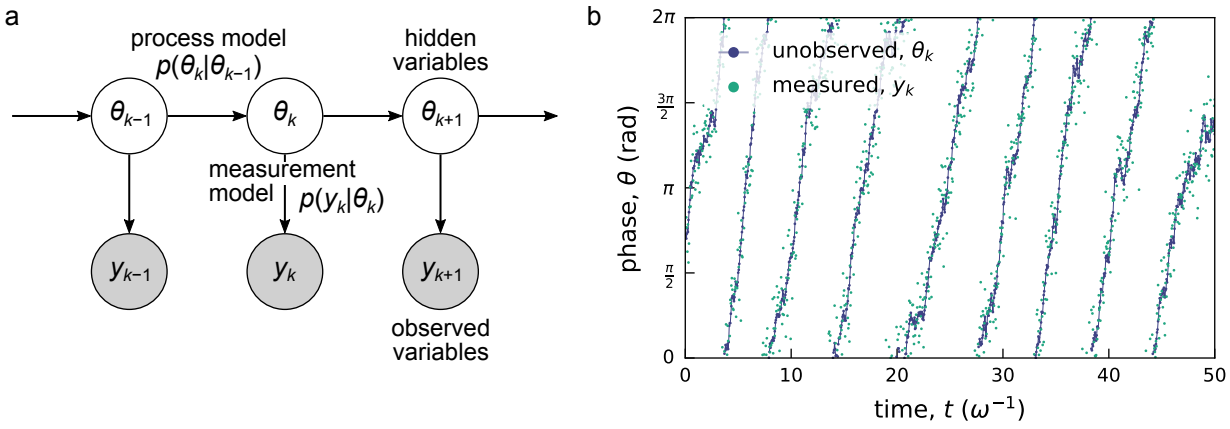


Figure S1: Phase drift and diffusion with measurement noise. (a) Graphical model showing the conditional dependencies among the hidden phase variables θ_k and the measured phase y_k for $k = 1, \ldots, N$. (b) Simulated data for the unobserved phase $\theta_{1:N}$ and the measured phase $y_{1:N}$ for $N = 2000$ time steps. We measure time in units of ω^{-1} and phase in radians from an origin at θ_0. The three dimensionless parameters are the time step $\Delta t = 0.05\omega^{-1}$, the phase diffusivity $D_\theta = 0.1\omega$, and the measurement noise $R = 0.2$.

$\theta_{k-1}, \omega, D_\theta)$, that describes the dynamics of the latent phase variable, and (2) the **measurement model**, $p(y_k \mid \theta_k, R)$, that describes the relation between the observed variables and the latent variables. Here, the process model is

$$p(\theta_k \mid \theta_{k-1}, \omega, D_\theta) = \mathcal{N}(\theta_k \mid \theta_{k-1} + \omega\Delta t, 2D_\theta\Delta t) \tag{2.4}$$

and the measurement model is

$$p(y_k \mid \theta_k, R) = \mathcal{N}(y_k \mid \theta_k, R) \tag{2.5}$$

Additionally, we specify a normally distributed prior for the initial phase θ_1 with mean μ_θ and variance σ_θ^2

$$p(\theta_1) = \mathcal{N}(\theta_1 \mid \mu_\theta, \sigma_\theta^2) \tag{2.6}$$

Figure S1b shows simulated data generated from this model with frequency $\omega = 1$, time step $\Delta t = 0.05$, phase diffusivity $D_\theta = 0.1$, noise magnitude $R = 0.2$.

2.2 Inferring the Latent Phase, $p(\theta_{1:N} \mid y_{1:N}, \lambda)$

Given the model parameters $\lambda = \{\omega, D_\theta, R\}$ and the noisy measurements, the posterior distribution over latent phase trajectories is

$$p(\theta_{1:N} \mid y_{1:N}, \lambda) = \frac{p(\theta_{1:N}, y_{1:N} \mid \lambda)}{p(y_{1:N} \mid \lambda)} \tag{2.7}$$

which is proportional to the complete-data likelihood of equation (2.3). Because both the process model and the measurement model are Gaussian, the smoothing distribution is multivariate normal

$$p(\theta_{1:N} \mid y_{1:N}, \lambda) = \mathcal{N}\Big(\theta_{1:N} \mid \hat{\theta}_{1:N}, \Sigma\Big) \tag{2.8}$$

The posterior mean $\hat{\theta}_{1:N}$ and covariance Σ are computed in $\mathcal{O}(N)$ time using a forward Kalman filter followed by Rauch-Tung-Striebel (RTS) smoothing.[6,7]

2.2.1 Forward Kalman Filter, $p(\theta_k \mid y_{1:k}, \lambda)$

The Kalman filter recursively computes the posterior distribution of the latent phase θ_k conditioned on observations up to time k, denoted $p(\theta_k \mid y_{1:k})$, where $y_{1:k} = \{y_1, \ldots, y_k\}$. For notational simplicity, we omit explicit conditioning on the parameters λ. Because the model is linear and Gaussian, all filtering distributions are Gaussian,

$$p(\theta_k \mid y_{1:k}) = \mathcal{N}(\theta_k \mid \hat{\theta}_{k|k}, P_{k|k}), \tag{2.9}$$

where $\hat{\theta}_{k|k}$ and $P_{k|k}$ denote the posterior mean and variance, respectively. The recursion from step $k-1$ to k consists of two stages: a **prediction** step and an **update** step.

Prediction. Before observing y_k, we use the process model to predict the unobserved phase θ_k conditioned on the prior observations $y_{1:k-1}$ as

$$\begin{aligned} p(\theta_k \mid y_{1:k-1}) &= \int p(\theta_k \mid \theta_{k-1}) p(\theta_{k-1} \mid y_{1:k-1}) d\theta_{k-1} \\ &= \mathcal{N}(\theta_k \mid \hat{\theta}_{k|k-1}, P_{k|k-1}) \end{aligned} \tag{2.10}$$

The resulting distribution is Gaussian with mean and variance

$$\begin{aligned} \hat{\theta}_{k|k-1} &= \hat{\theta}_{k-1|k-1} + \omega \Delta t \\ P_{k|k-1} &= P_{k-1|k-1} + 2D_\theta \Delta t \end{aligned} \tag{2.11}$$

Update. Upon observing y_k, we use Bayes' rule to update our estimate for θ_k as

$$\begin{aligned} p(\theta_k \mid y_{1:k}) &\propto p(y_k \mid \theta_k) p(\theta_k \mid y_{1:k-1}) \\ &= \mathcal{N}(\theta_k \mid \hat{\theta}_{k|k}, P_{k|k}) \end{aligned} \tag{2.12}$$

The updated distribution remains Gaussian with mean and variance

$$\begin{aligned} \hat{\theta}_{k|k} &= \hat{\theta}_{k|k-1} + K_k\big(y_k - \hat{\theta}_{k|k-1}\big) \\ P_{k|k} &= (1 - K_k) P_{k|k-1} \end{aligned} \tag{2.13}$$

Here, the Kalman gain $K_k = P_{k|k-1}(P_{k|k-1} + R)^{-1}$ determines the relative weight given to the prediction and the measurement. Equivalently, the posterior precision is the sum of the prior precision $P_{k|k-1}^{-1}$ and the measurement precision R^{-1}, and the posterior mean is the precision-weighted average of the prediction $\hat{\theta}_{k|k-1}$ and the observation y_k.

2.2.2 Backward RTS Smoothing, $p(\theta_k \mid y_{1:N}, \lambda)$

During the smoothing step, we progress backwards through the latent phases θ_k from $k = N$ to $k = 1$, conditioning on all observations $y_{1:N}$. In progressing from index $k+1$ to k, we use the relationship

$$p(\theta_k \mid y_{1:N}) = p(\theta_k \mid y_{1:k}) \int \frac{p(\theta_{k+1} \mid \theta_k)\, p(\theta_{k+1} \mid y_{1:N})}{p(\theta_{k+1} \mid y_{1:k})}\, d\theta_{k+1} \tag{2.14}$$

which combines the filtered distribution at time k with information from future observations through the smoothed distribution at time $k+1$. The distributions appearing in Eq. (2.14) are

$$\begin{aligned} p(\theta_k \mid y_{1:k}) &= \mathcal{N}(\theta_k \mid \hat{\theta}_{k|k}, P_{k|k}) \\ p(\theta_{k+1} \mid \theta_k) &= \mathcal{N}(\theta_{k+1} \mid \theta_k + \omega \Delta t,\, 2D_\theta \Delta t) \\ p(\theta_{k+1} \mid y_{1:k}) &= \mathcal{N}(\theta_{k+1} \mid \hat{\theta}_{k+1|k}, P_{k+1|k}) \end{aligned} \tag{2.15}$$

Equation (2.14) follows from the conditional dependencies of the graphical model (Fig. S1a), which imply that

$$p(\theta_k \mid \theta_{k+1}, y_{1:N}) = p(\theta_k \mid \theta_{k+1}, y_{1:k}) \tag{2.16}$$

Because the process and measurement models are linear and Gaussian, the smoothing distribution is Gaussian

$$p(\theta_k \mid y_{1:N}) = \mathcal{N}(\theta_k \mid \hat{\theta}_{k|N}, P_{k|N}). \tag{2.17}$$

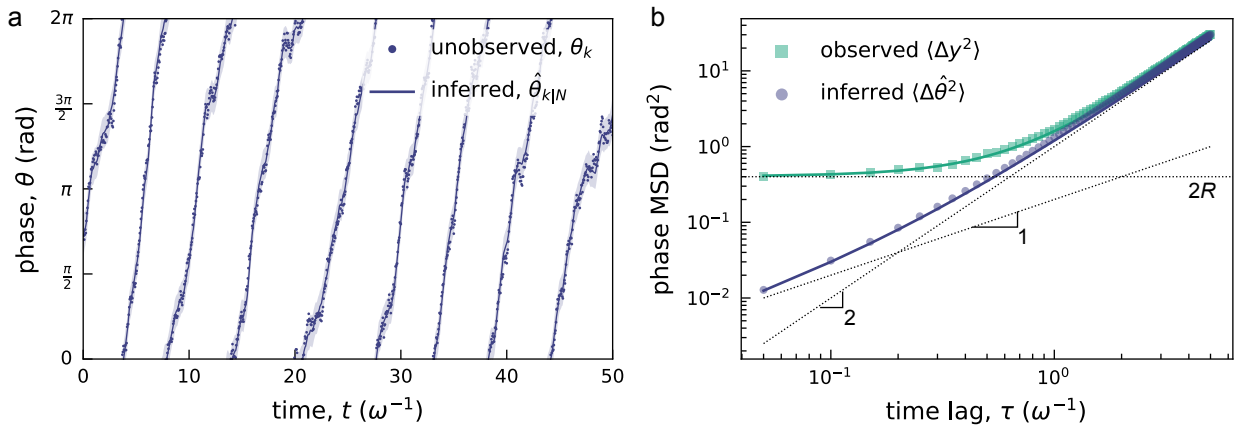


Figure S2: Inferred phases $p(\theta_{1:N} \mid y_{1:N}, \lambda)$. (a) Given the model parameters λ used to simulate the data, we infer the unobserved phases $\theta_{1:N}$ from the measured positions $y_{1:N}$ via Kalman filtering and smoothing. The solid curve denotes the posterior mean $\hat{\theta}_{k|N}$; the shaded region denotes the 95% credible interval. (b) Mean squared displacement (MSD) vs. lag time τ based on the measured phases $y_{1:N}$ and the inferred phases $\hat{\theta}_{1:N}$ (markers). Solid curves denote the theoretical expectation with measurement noise, $2\sigma^2 + 2D_\theta\tau + (\omega\tau)^2$, and without measurement noise, $2D_\theta\tau + (\omega\tau)^2$. The dashed curves show the respective contributions of measurement noise $2\sigma^2$, phase diffusion $2D_\theta\tau$, and phase drift $\omega\tau$.

with mean and variance computed using the Rauch–Tung–Striebel recursion relations

$$\begin{aligned} \hat{\theta}_{k|N} &= \hat{\theta}_{k|k} + C_k\big(\hat{\theta}_{k+1|N} - \hat{\theta}_{k+1|k}\big) \\ P_{k|N} &= P_{k|k} + C_k\big(P_{k+1|N} - P_{k+1|k}\big)C_k^\top \end{aligned} \tag{2.18}$$

Here, the RTS smoothing gain $C_k = P_{k|k}P_{k+1|k}^{-1}$ controls how strongly future observations modify the filtered estimate at time $\hat{\theta}_{k|k}$, which itself depends only on past measurements.

2.2.3 Posterior Covariance, Σ

The posterior distribution $p(\theta_{1:N} \mid y_{1:N}, \lambda)$ is multivariate normal with mean $\hat{\theta}_{1:N} = \{\hat{\theta}_{1|N}, \ldots, \hat{\theta}_{N|N}\}$ given by Kalman filtering and RTS smoothing. We now determine its covariance matrix Σ.

Full matrix. The log-posterior for the latent phase trajectory is

$$\log p(\theta_{1:N} \mid y_{1:N}, \lambda) = \text{const.} - \frac{(\theta_1 - \mu_\theta)^2}{2\sigma_\theta^2} - \sum_{k=2}^{N} \frac{(\theta_k - \theta_{k-1} - \omega\Delta t)^2}{4D_\theta\Delta t} - \sum_{k=1}^{N} \frac{(y_k - \theta_k)^2}{2R} \tag{2.19}$$

where μ_θ and σ_θ^2 are fixed hyperparameters of a Gaussian prior on the initial phase. Because this expression is quadratic in $\theta_{1:N}$, the posterior precision matrix Σ^{-1} is given by the negative Hessian

$$\Sigma_{j,l}^{-1} = -\nabla_{\theta_j}\nabla_{\theta_l} \log p(\theta_{1:N} \mid y_{1:N}, \lambda). \tag{2.20}$$

For interior indices $j = 2, \ldots, N-1$, this yields the tridiagonal structure

$$\Sigma_{j,l}^{-1} = -\frac{1}{2D_\theta\Delta t}\,\delta_{l,j-1} + \left(\frac{1}{D_\theta\Delta t} + \frac{1}{R}\right)\delta_{l,j} - \frac{1}{2D_\theta\Delta t}\,\delta_{l,j+1}, \tag{2.21}$$

The boundary diagonal entries are

$$\Sigma_{1,1}^{-1} = \frac{1}{2D_\theta \Delta t} + \frac{1}{R} + \frac{1}{\sigma_\theta^2}, \qquad \Sigma_{N,N}^{-1} = \frac{1}{2D_\theta \Delta t} + \frac{1}{R} \tag{2.22}$$

This band-diagonal matrix can be inverted in $\mathcal{O}(N)$ time using Thomas' algorithm (e.g., SciPy's `solve_banded` function) to obtain the full covariance matrix Σ. The diagonal entries $\Sigma_{k,k}$ coincide with the smoothed variances $P_{k|N}$ obtained from the RTS smoother.

Diagonal and lag-one entries. For evaluating expectations in the EM updates, we require only the marginal variances $\Sigma_{k,k} = P_{k|N}$ and the lag-one covariances $\Sigma_{k,k-1} = \mathrm{Cov}(\theta_k, \theta_{k-1} \mid y_{1:N})$. Because the posterior is Gaussian, the pairwise joint distribution $p(\theta_{k-1}, \theta_k \mid y_{1:N})$ is also Gaussian. The Rauch–Tung–Striebel recursion implies that the lag-one covariance satisfies

$$\Sigma_{k,k-1} = C_{k-1} P_{k|N}, \tag{2.23}$$

where $C_{k-1} = P_{k-1|k-1}(P_{k|k-1})^{-1}$ is the RTS smoothing gain. This identity follows from propagating covariances through the linear smoothing update (2.18). Thus, both the diagonal entries $\Sigma_{k,k}$ and the lag-one entries $\Sigma_{k,k-1}$ are obtained directly from the smoothing recursion without constructing or inverting the full covariance matrix.

2.2.4 Marginal Likelihood, $p(y_{1:N} \mid \lambda)$

In addition to inferring latent phase trajectories, the Kalman filter also provides a convenient method for evaluating the marginal likelihood, obtained by integrating out the latent phases from the complete-data likelihood

$$p(y_{1:N} \mid \lambda) = \int p(\theta_{1:N}, y_{1:N} \mid \lambda) d\theta_{1:N} \tag{2.24}$$

This quantity will be used below in estimating the model parameters $\lambda = \{\omega, D_\theta, R\}$. The marginal likelihood can be factored using the chain rule as

$$p(y_{1:N} \mid \lambda) = \prod_{k=1}^{N} p(y_k \mid y_{1:k-1}, \lambda) \tag{2.25}$$

Here, the predictive distribution of observation y_k given past measurements $y_{1:k-1}$ is given by the Kalman prediction and the measurement model as

$$p(y_k \mid y_{1:k-1}) = \mathcal{N}(y_k \mid \hat{\theta}_{k|k-1}, P_{k|k-1} + R) \tag{2.26}$$

Thus the log marginal likelihood is

$$\log p(y_{1:N} \mid \lambda) = -\frac{1}{2} \sum_{k=1}^{N} \left[\log\left(2\pi(P_{k|k-1} + R)\right) + \frac{(y_k - \hat{\theta}_{k|k-1})^2}{P_{k|k-1} + R} \right] \tag{2.27}$$

2.2.5 Sampling Phase Trajectories

We can sample phase trajectories from the smoothing distribution $p(\theta_{1:N} \mid y_{1:N}, \lambda)$ using the Forward-Filter Backward-Sampling (FFBS) algorithm. These sampled trajectories are useful for model criticism and for computing expectations that arise in uncertainty quantification of the parameters λ. We first sample θ_N from $p(\theta_N \mid y_{1:N}, \lambda)$, which has mean $\hat{\theta}_{N|N}$ and variance $P_{N|N}$. At the final time step, the filtered and smoothed distributions coincide. Proceeding backwards for $k = N, \ldots, 2$, we sample θ_{k-1} conditioned on θ_k from

$$p(\theta_{k-1} \mid \theta_k, y_{1:N}, \lambda) = p(\theta_{k-1} \mid \theta_k, y_{1:k-1}, \lambda) \propto p(\theta_k \mid \theta_{k-1}, y_{1:k-1}) p(\theta_{k-1} \mid y_{1:k-1}), \tag{2.28}$$

since conditioning on θ_k removes dependence on future observations $y_{k+1:N}$. Because these distributions are Gaussian, the conditional distribution is Gaussian

$$\theta_{k-1} \mid \theta_k, y_{1:N} \sim \mathcal{N}\Big(\hat{\theta}_{k-1|k-1} + C_{k-1}(\theta_k - \hat{\theta}_{k|k-1}),\ P_{k-1|k-1} - C_{k-1} P_{k|k-1} C_{k-1}^\top\Big) \tag{2.29}$$

where $C_{k-1} = P_{k-1|k-1}(P_{k|k-1})^{-1}$ is the RTS smoothing gain. This form applies in the present model because the latent phase evolves linearly as equation (2.1), so the backward-sampling gain coincides with the smoothing gain.

2.2.6 Mean Squared Displacement

The mean squared displacement (MSD) of the noisy observations $y_{1:N}$ deviates significantly from the noise-free expectation $2D_\theta t + (\omega t)^2$ (Fig. S2b). Having inferred the latent phases, we can evaluate the MSD under the smoothing distribution $p(\theta_{1:N} \mid y_{1:N}, \lambda)$. For a fixed lag n, we compute

$$\mathbb{E}\big[(\theta_{k+n} - \theta_k)^2 \mid y_{1:N}, \lambda\big] = (\hat{\theta}_{k+n|N} - \hat{\theta}_{k|N})^2 + \Sigma_{k+n,k+n} + \Sigma_{k,k} - 2\Sigma_{k+n,k} \tag{2.30}$$

where $\Sigma_{k,j}$ denotes the posterior covariance between θ_k and θ_j. The MSD is then estimated as

$$\mathrm{MSD}(n) = \frac{1}{N-n} \sum_{k=1}^{N-n} \mathbb{E}\big[(\theta_{k+n} - \theta_k)^2 \mid y_{1:N}, \lambda\big], \tag{2.31}$$

with lag time $t_n = n\Delta t$. Figure S2b shows that the inferred MSD is in excellent agreement with the theoretical expectation $\mathrm{MSD}(n) = 2D_\theta t_n + (\omega t_n)^2$. The above analysis assumes perfect knowledge of the model parameters λ used to generate the data. In practice, these parameters—namely the phase velocity ω, the phase diffusivity D_θ, the measurement noise magnitude R—are unknown and must be inferred from the observed data.

2.3 Estimating Parameters, $p(\lambda \mid y_{1:N})$

We now seek to infer the model parameters $\lambda = \{\omega, D_\theta, R\}$ given the noisy phase measurements y (short for $y_{1:N}$). The posterior distribution is

$$p(\lambda \mid y) \propto p(\lambda) p(y \mid \lambda) \tag{2.32}$$

where the marginal likelihood $p(y \mid \lambda)$ of equation (2.27) is provided by the Kalman filter, and the prior $p(\lambda)$ is defined below. Up to an additive constant, the log posterior is

$$\mathcal{L}(\lambda) = \log p(\lambda) + \log p(\lambda \mid y) \tag{2.33}$$

Maximizing this function with respect to λ gives the maximum a posterior (MAP) parameter estimates

$$\hat{\lambda} = \arg\max_{\lambda} \mathcal{L}(\lambda) \tag{2.34}$$

Expanding $\mathcal{L}(\lambda)$ in a Taylor series about the MAP estimate $\hat{\lambda}$ gives

$$\mathcal{L}(\lambda) = \mathcal{L}(\hat{\lambda}) - \frac{1}{2}(\lambda - \hat{\lambda})^\top \mathcal{I}(\hat{\lambda})(\lambda - \hat{\lambda}) + \dots \tag{2.35}$$

where $\mathcal{I}(\hat{\lambda}) = -\nabla^2_\lambda \mathcal{L}(\lambda)|_{\hat{\lambda}}$ is the posterior information matrix characterizing the local curvature of the log posterior. Under this approximation—the Laplace or normal approximation—the posterior distribution is approximated as a multivariate normal distribution with mean $\hat{\lambda}$ and precision matrix $\mathcal{I}(\hat{\lambda})$. Below, we discuss different strategies for evaluating these two key features of the posterior: the MAP estimate $\hat{\lambda}$ and the posterior information matrix $\mathcal{I}(\hat{\lambda})$.

2.3.1 Prior Distribution, $p(\lambda)$

To incorporate prior information and to regularize parameter estimation, we place prior distributions on the unknown parameters $\lambda = \{\omega, D_\theta, R\}$. For mathematical convenience, we select priors that are conjugate to the *complete-data likelihood* $p(\theta, y \mid \lambda)$, which simplifies both the EM updates and the analytic evaluation of posterior curvature discussed below. Specifically, we assign a normal prior to the frequency and inverse-gamma priors to the phase diffusivity and measurement-noise variance:

$$\omega \sim \mathcal{N}(\mu_\omega, \sigma_\omega^2), \quad D_\theta \sim \text{Inv-Gamma}(\alpha_D, \beta_D), \quad R \sim \text{Inv-Gamma}(\alpha_R, \beta_R) \tag{2.36}$$

The hyperparameters α and β for the inverse-gamma distributions can be chosen to specify the mean $\beta/(\alpha-1)$ and variance $\beta^2/[(\alpha-1)^2(\alpha-2)]$ of the associated parameters. To express ignorance in the model parameters, one may take $\sigma_\omega \to \infty$, $\alpha \to 0$, and $\beta \to 0$, yielding the improper limits

$$p(\omega) \propto \text{constant}, \quad p(D_\theta) \propto \frac{1}{D_\theta}, \quad p(R) \propto \frac{1}{R} \tag{2.37}$$

2.3.2 MAP Estimation, $\hat{\lambda}$

We consider three complementary strategies for computing the maximum a posteriori (MAP) estimate $\hat{\lambda}$. First, gradient-free optimization directly maximizes the log posterior using only function evaluations and is simple and robust in low dimensions. Second, gradient-based optimization leverages the analytic posterior gradient obtained via the Louis identity, enabling efficient quasi-Newton updates with rapid convergence near the optimum. Third, the expectation–maximization (EM) algorithm alternates between smoothing and closed-form parameter updates under conjugate priors, yielding monotonic ascent of the posterior. In practice, supplying the analytic gradient to a quasi-Newton method (e.g., BFGS) provides the best computational performance, while EM offers interpretability and guaranteed ascent but typically converges more slowly.

Gradient-free optimization. A straightforward approach to MAP estimation is to maximize the log posterior $\mathcal{L}(\lambda)$ using gradient-free optimization methods. These algorithms require only evaluations of $\mathcal{L}(\lambda)$, making them simple to implement and robust when analytic derivatives are unavailable or unreliable. In practice, we perform the optimization over $\{\omega, \log D_\theta, \log R\}$ to to enforce positivity of D_θ and R and to improve numerical conditioning, while the posterior itself remains defined in the original parameters. Among the derivative-free methods available in `scipy.optimize`, we find Powell's method to converge reliably and efficiently in this low-dimensional problem. The Nelder–Mead simplex method yields similar results but typically requires more function evaluations and converges more slowly near the optimum.

Gradient-based optimization. More efficient optimization can be achieved by supplying the gradient of the log posterior to a quasi-Newton method such as BFGS. The posterior gradient is

$$\nabla_\lambda \mathcal{L}(\lambda) = \nabla_\lambda \log p(\lambda) + \nabla_\lambda \log p(y \mid \lambda) \tag{2.38}$$

where the first gradient is analytic and the second—also called the score—involves the marginal likelihood (2.27). Direct differentiation of the marginal log likelihood is inconvenient because it requires differentiating through the Kalman filter recursions. Instead, we use the Louis identity,[8] which expresses derivatives of the marginal log likelihood in terms of the complete-data log likelihood $\ell_c(\lambda; \theta, y) = \log p(\theta, y \mid \lambda)$ as

$$\nabla_\lambda \log p(y \mid \lambda) = \mathbb{E}[\nabla_\lambda \ell_c(\lambda; \theta, y) \mid y, \lambda] \tag{2.39}$$

where the expectation is taken with respect to the smoothing distribution $p(\theta \mid y, \lambda)$ provided by the Kalman filter with Rauch–Tung–Striebel smoothing.

For the present problem, gradients of the complete-data likelihood $\ell_c(\lambda; \theta, y)$ with respect to $\{\omega, \log D_\theta, \log R\}$ are given by

$$\begin{aligned}
\partial_\omega \ell_c &= \frac{S_1 - (N-1)\omega\Delta t}{2D_\theta} \\
\partial_{\log D_\theta} \ell_c = D_\theta \partial_{D_\theta} \ell_c &= -\frac{(N-1)}{2} + \frac{S_0 - 2S_1\omega\Delta t + (N-1)(\omega\Delta t)^2}{4D_\theta \Delta t} \\
\partial_{\log R} \ell_c = R\, \partial_R \ell_c &= -\frac{N}{2} + \frac{S_y}{2R}
\end{aligned} \tag{2.40}$$

where the sufficient statistics S_0, S_1, and S_y are

$$S_0 = \sum_{k=2}^{N} (\theta_k - \theta_{k-1})^2, \quad S_1 = \sum_{k=2}^{N} \theta_k - \theta_{k-1} = \theta_N - \theta_1, \quad S_y = \sum_{k=1}^{N} (y_k - \theta_k)^2 \tag{2.41}$$

The expectations of these quantities under the smoothing distribution are

$$\begin{aligned}
\mathbb{E}[S_0 \mid y, \lambda] &= \sum_{k=2}^{N} \Big[(\hat\theta_k - \hat\theta_{k-1})^2 + \Sigma_{k,k} + \Sigma_{k-1,k-1} - 2\Sigma_{k,k-1} \Big] \\
\mathbb{E}[S_1 \mid y, \lambda] &= \hat\theta_N - \hat\theta_1 \\
\mathbb{E}[S_y \mid y, \lambda] &= \sum_{k=1}^{N} \Big[(y_k - \hat\theta_k)^2 + \Sigma_{k,k} \Big]
\end{aligned} \tag{2.42}$$

where $\hat{\theta}_k = \mathbb{E}[\theta_k \mid y, \lambda]$ denotes the smoothed mean and $\Sigma_{i,j} = \mathrm{Cov}(\theta_i, \theta_j \mid y, \lambda)$ the smoothed covariance. Substituting these expectations into the Louis identity yields the score $\nabla_\lambda \log p(y \mid \lambda)$, which we combine with $\nabla_\lambda \log p(\lambda)$ to obtain the full posterior gradient $\nabla_\lambda \mathcal{L}(\lambda)$. Providing this analytic score substantially accelerates convergence relative to gradient-free methods, particularly near the optimum where quasi-Newton updates exhibit superlinear convergence.

Expectation Maximization (EM) algorithm. Alternatively, the MAP estimates $\hat{\lambda}$ can be computed through repeated iterations of "expectation" and "maximization". In the **E-step**, we compute the smoothing distribution $p(\theta \mid y, \lambda^{(j)})$ using the Kalman filter with Rauch–Tung–Striebel smoothing for the current parameter estimates $\lambda^{(j)}$. From the smoothed means $\hat{\theta}_k$ and covariances $\Sigma_{i,j}$, we evaluate the expected sufficient statistics

$$\mathbb{E}[S_0 \mid y, \lambda^{(m)}], \qquad \mathbb{E}[S_1 \mid y, \lambda^{(m)}], \qquad \mathbb{E}[S_y \mid y, \lambda^{(m)}] \tag{2.43}$$

In the subsequent **M-step**, we maximize the expectation of the log posterior with respect to the smoothing distribution

$$Q(\lambda \mid \lambda^{(m)}) = \mathbb{E}\Big[\mathcal{L}(\lambda) \mid y, \lambda^{(m)}\Big] = \mathbb{E}\Big[\ell_c(\lambda; \theta, y) \mid y, \lambda^{(m)}\Big] + \log p(\lambda) \tag{2.44}$$

Under conjugate priors, Q separates over parameters and admits closed-form updates.

With a Gaussian prior $\omega \sim \mathcal{N}(\mu_\omega, \sigma_\omega^2)$, conditional maximization yields

$$\omega^{(m+1)} = \left(\frac{\mathbb{E}[S_1]}{2D_\theta^{(m)}} + \frac{\mu_\omega}{\sigma_\omega^2}\right)\left(\frac{(N-1)\Delta t}{2D_\theta^{(m)}} + \frac{1}{\sigma_\omega^2}\right)^{-1} \tag{2.45}$$

This expression can be interpreted as a weighted average of the data-driven estimate $\mathbb{E}[S_1]/(N-1)\Delta t$ and the prior mean μ_ω, with weights determined by the relative precisions of the trajectory and the prior. With an inverse-gamma prior $D_\theta \sim \text{Inv-Gamma}(\alpha_D, \beta_D)$, the update is

$$D_\theta^{(m+1)} = \frac{\mathbb{E}[S_0] - 2\mathbb{E}[S_1]\omega^{(m+1)}\Delta t + (N-1)(\omega^{(m+1)}\Delta t)^2 + 4\beta_D \Delta t}{2(N - 1 + 2(\alpha_D + 1))\,\Delta t} \tag{2.46}$$

Finally, with $R \sim \text{Inv-Gamma}(\alpha_R, \beta_R)$, the noise variance can be updated as

$$R^{(m+1)} = \frac{\mathbb{E}[S_y] + 2\beta_R}{N + 2(\alpha_R + 1)} \tag{2.47}$$

Each iteration increases $Q(\lambda \mid \lambda^{(m)})$ and therefore monotonically increases the posterior $\mathcal{L}(\lambda)$. At convergence, the EM iterates satisfy $\lambda^{(m)} \to \hat{\lambda}$, where $\hat{\lambda}$ is a fixed point of the EM update:

$$\hat{\lambda} = \arg\max_\lambda Q(\lambda \mid \hat{\lambda}) \tag{2.48}$$

The first-order optimality condition implies

$$\nabla_\lambda Q(\lambda \mid \hat{\lambda})\big|_{\lambda=\hat{\lambda}} = \Big(\mathbb{E}[\nabla_\lambda \ell_c(\lambda; \theta, y) \mid y, \hat{\lambda}] + \nabla_\lambda \log p(\lambda)\Big)_{\lambda=\hat{\lambda}} = 0 \tag{2.49}$$

which, by the Louis identity (2.39), is equivalent to $\nabla_\lambda \mathcal{L}(\lambda)\big|_{\lambda=\hat{\lambda}} = 0$, demonstrating that EM and gradient-based optimization target the same stationary points of the posterior.

2.3.3 Posterior Uncertainty via Laplace Approximation, $\mathcal{I}(\hat{\lambda})$

Uncertainty in the MAP estimate $\hat{\lambda}$ is quantified using a Laplace approximation to the posterior. Expanding the log posterior $\mathcal{L}(\lambda)$ to second order about $\hat{\lambda}$ yields a local Gaussian approximation

$$p(\lambda \mid y) \approx \mathcal{N}\Big(\hat{\lambda}, \mathcal{I}^{-1}(\hat{\lambda})\Big) \tag{2.50}$$

where the posterior information matrix is the negative Hessian of the log posterior evaluated at the optimum,

$$\mathcal{I}(\hat{\lambda}) = -\nabla^2_\lambda \mathcal{L}(\lambda)\big|_{\lambda=\hat{\lambda}}. \tag{2.51}$$

The diagonal entries of $\mathcal{I}^{-1}$ provide approximate posterior variances, while the full matrix captures parameter correlations. We consider both numerical and analytical strategies for evaluating $\mathcal{I}(\hat{\lambda})$.

Numerical Hessian. The most direct approach is to compute the Hessian of $\mathcal{L}(\lambda)$ by finite differences of either the log posterior itself or its gradient. In practice, central-difference schemes applied to the analytic score $\nabla_\lambda \mathcal{L}(\lambda)$ provide improved stability and second-order accuracy. This approach is straightforward to implement and makes minimal use of model structure, but it requires multiple evaluations of the Kalman smoother and can become computationally expensive as the parameter dimension grows.

Analytical decomposition via Louis identity. An alternative exploits the Louis identity to express the observed information in terms of complete-data quantities. Differentiating the Louis identity (2.39) with respect to the parameters λ, we obtain the decomposition

$$\underbrace{-\nabla^2_\lambda \log p(y \mid \lambda)}_{\mathcal{I}_{\text{like}}(\lambda)} = \underbrace{-\mathbb{E}\big[\nabla^2_\lambda \ell_c(\lambda;\theta,y) \mid y,\lambda\big]}_{\mathcal{I}_{\text{comp}}(\lambda)} - \underbrace{\text{Cov}(\nabla_\lambda \ell_c(\lambda;\theta,y) \mid y,\lambda)}_{\mathcal{I}_{\text{miss}}(\lambda)} \tag{2.52}$$

where $\mathcal{I}_{\text{comp}}(\lambda)$ represents the information that would be available if the latent phases were observed (complete-data information), while $\mathcal{I}_{\text{miss}}(\lambda)$ quantifies the information lost due to latent-state uncertainty (missing information). Including the prior contribution, the full posterior information matrix becomes

$$\begin{aligned} \mathcal{I}(\lambda) &= \mathcal{I}_{\text{prior}}(\lambda) + \mathcal{I}_{\text{like}}(\lambda) \\ &= \mathcal{I}_{\text{prior}}(\lambda) + \mathcal{I}_{\text{comp}}(\lambda) - \mathcal{I}_{\text{miss}}(\lambda) \end{aligned} \quad \text{with} \quad \mathcal{I}_{\text{prior}} = -\nabla^2_\lambda \log p(\lambda), \tag{2.53}$$

This decomposition provides both computational and conceptual advantages: it avoids differentiating through the Kalman recursions and clarifies how uncertainty in the latent phase reduces parameter information. Moreover, $\mathcal{I}_{\text{miss}}$ vanishes in the limit of negligible measurement noise, recovering the complete-data information as expected.

Notably, the missing information is absent from the EM objective function $Q(\lambda \mid \lambda^{(m)})$ of equation (2.44). At convergence $\lambda^{(m)} \to \hat{\lambda}$, the negative hessian of the EM objective is

$$\begin{aligned} -\nabla^2_\lambda Q(\lambda \mid \hat{\lambda})\big|_{\lambda=\hat{\lambda}} &= \Big(-\mathbb{E}[\nabla^2_\lambda \ell_c(\lambda;\theta,y) \mid y,\hat{\lambda}] - \nabla^2_\lambda \log p(\lambda)\Big)_{\lambda=\hat{\lambda}} \\ &= \mathcal{I}_{\text{comp}}(\hat{\lambda}) + \mathcal{I}_{\text{prior}}(\hat{\lambda}) \end{aligned} \tag{2.54}$$

The missing-information term precisely quantifies the discrepancy between the curvature of the EM surrogate objective and that of the true posterior.

2.4 Quantifying Performance

To study the precision and accuracy of the inference procedure, we generate synthetic data sets using the generative model described above. The parameters used to simulate the data are denoted with a superscript "true". For each synthetic data set we then infer the model parameters $\lambda = (\omega, \log D_\theta, \log R)$ and compute the maximum a posteriori (MAP) estimates $\hat{\lambda}$ together with Laplace error bars obtained from the Gaussian approximation to the posterior distribution.

Figure S3 summarizes the inferred parameters for the synthetic phase data $y_{1:N}$ shown in Figure S1b. Panel (a) shows the marginal posterior distributions for each parameter, while panel (b) shows the joint marginal distributions for parameter pairs using the Laplace approximation. The posterior distributions are centered close to the true parameter values used to generate the data, indicating that the inference procedure recovers the underlying dynamics accurately for this trajectory. The contour plots further reveal correlations between parameters that arise from the structure of the model, particularly between the diffusivity D_θ and the measurement noise variance R, which both contribute to fluctuations in the observed phase increments.

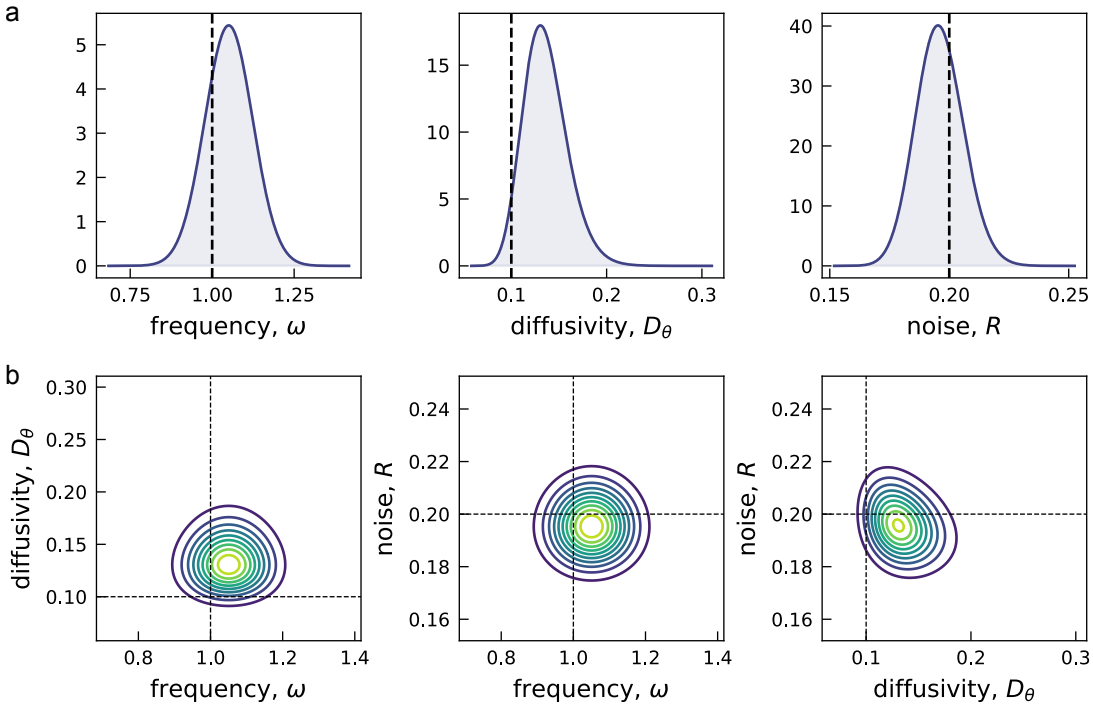


Figure S3: (a) Marginal posterior distributions for the frequency ω, phase diffusivity D_θ, and noise variance R given the synthetic phase data $y_{1:N}$ shown in Figure S1b. Dashed lines indicate the true parameter values used to generate the data. (b) Contour plots of the joint marginal posterior distributions for parameter pairs, estimated using the Laplace approximation in the log-transformed parameterization $\lambda = \{\omega, \log D_\theta, \log R\}$.

2.4.1 Effects of N, Δt, and R

To understand how experimental design influences parameter estimation, we systematically vary the number of measurements N, the sampling interval Δt, and the measurement noise variance R^{true} in synthetic data sets. Without loss of generality, we set $\omega^{\text{true}} = 1$ and $D_\theta^{\text{true}} = 1$, which defines

the natural units of time and phase. With this choice, time is measured in units of $D_\theta^{\mathrm{true}}/\omega_{\mathrm{true}}^2$ and phase in units of $(D_\theta^{\mathrm{true}}/\omega_{\mathrm{true}})^{1/2}$. The remaining parameters controlling the inference problem are the number of measurements N, the sampling interval Δt, and the measurement noise variance R^{true} used to generate the data. For each realization we compute the MAP estimate together with a Laplace error bar, defined as the standard deviation obtained from the marginal normal approximation to the posterior distribution.

Figure S4a shows the effect of N on the parameter estimates for a time step $\Delta t = 1$ and noise variance $R^{\mathrm{true}} = 0.2$. The results are consistent with theoretical expectations for the zero-noise limit ($R \to 0$), in which the frequency and log-diffusivity estimates have variances

$$\mathrm{Var}(\omega) \approx \frac{2\hat{D}_\theta}{N\Delta t}, \qquad \mathrm{Var}(\log D_\theta) \approx \frac{2}{N}. \tag{2.55}$$

The shaded regions show the theoretical one–standard–deviation range predicted by these expressions. This low-noise approximation is valid when the variance of a single phase increment is dominated by phase diffusion,

$$\mathrm{Var}(y_k - y_{k-1}) = 2D_\theta \Delta t + 2R, \tag{2.56}$$

so that $R \ll D_\theta \Delta t$. In this limit, the noise variance R is difficult to infer from short trajectories because fluctuations in the observed phase increments arise from both phase diffusion and measurement noise. As a result, the posterior for R is dominated by the inverse-gamma prior when N is small, but the estimates converge toward the true value as N increases.

Figure S4b shows the effect of Δt on the parameter estimates for $N = 1000$ measurements with noise variance $R^{\mathrm{true}} = 0.2$. The uncertainty in the frequency is again well approximated by eqn. (2.55), which decreases inversely with the trajectory length $N\Delta t$. By contrast, the uncertainty in the inferred diffusivity is largely independent of the time step Δt, particularly in the low-noise regime where $D_\theta \Delta t \gg R$. In this regime the variance of the observed phase increments is dominated by diffusion rather than measurement noise, making the contribution of R difficult to identify. Consequently, estimates of the noise variance become highly uncertain for large time steps but are well resolved when $\Delta t \ll R/D_\theta$.

Figure S4c shows the effect of the true noise variance R^{true} on the parameter estimates for $N = 1000$ measurements with time step $\Delta t = 1$. Notably, the frequency is well resolved even for very noisy measurements with $R^{\mathrm{true}} = 10^5$. By contrast, the phase diffusivity D_θ becomes increasingly uncertain as the measurement noise grows and eventually dominates the variance of the observed phase increments ($R^{\mathrm{true}} \gg D_\theta \Delta t$). In this regime the fluctuations in the observations are governed primarily by measurement noise rather than phase diffusion, allowing the noise variance R to be estimated with high precision.

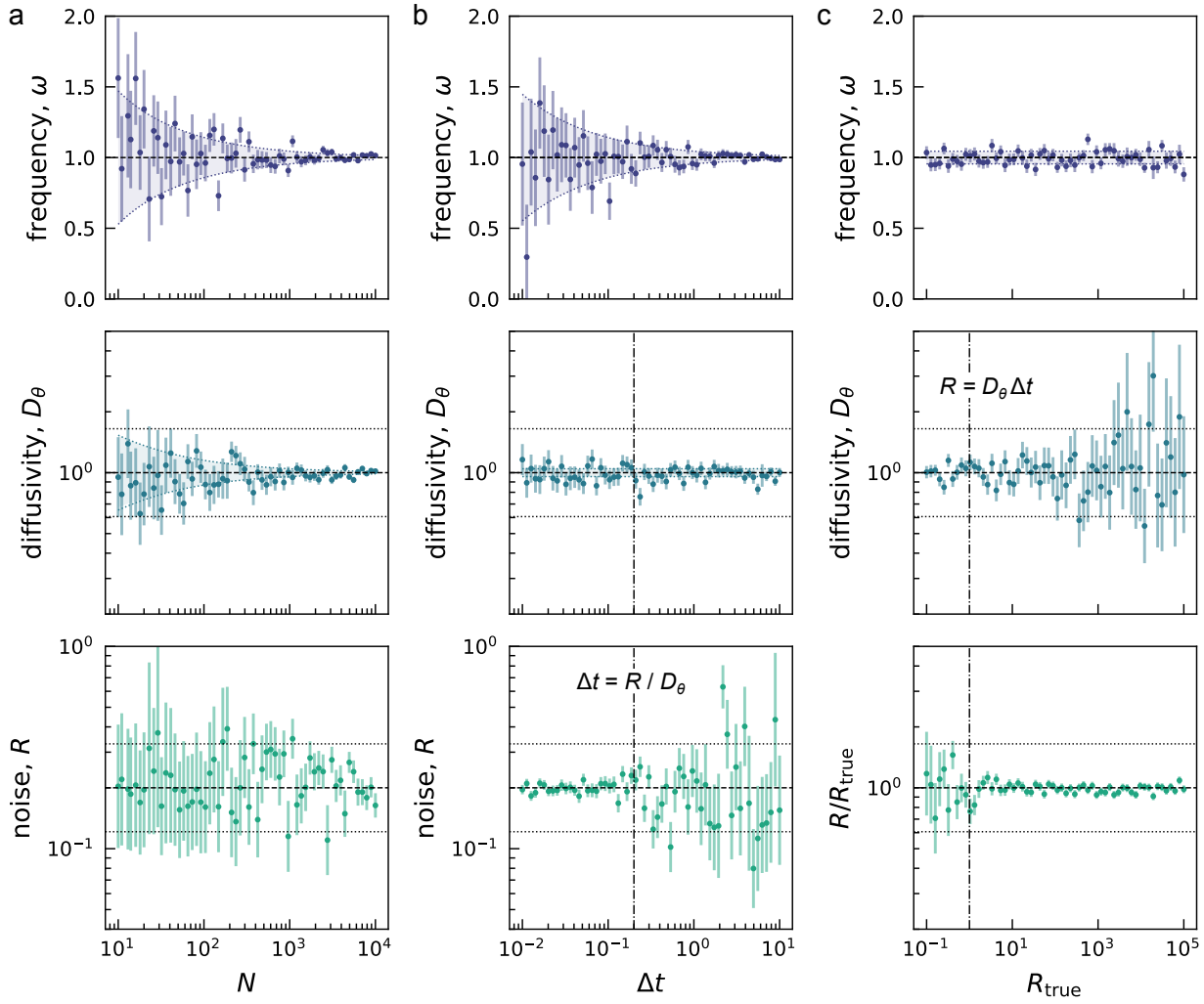


Figure S4: MAP estimates (markers) for the unknown frequency ω, phase diffusivity D_θ, and noise variance R as a function of (a) measurement count N, (b) time step Δt, and (c) true noise variance R^{true} used to generate the data. The other true values, $\omega^{\text{true}} = 1$ and $D_\theta^{\text{true}} = 1$, are indicated by black dashed lines. Error bars denote one posterior standard deviation obtained from the Laplace approximation. The priors for the frequency and diffusivity are $\mu_\omega = \sigma_\omega = 1$ and $\alpha_D = \beta_D = 2$. In (a), $\Delta t = 1$ and $R^{\text{true}} = 0.2$ with noise hyperparameters $\alpha_R = 2$ and $\beta_R = 0.4$. In (b), $N = 1000$ and $R^{\text{true}} = 0.2$ with $\alpha_R = 2$ and $\beta_R = 0.4$. In (c), $N = 1000$ and $\Delta t = 1$ with $\alpha_R = 2$ and $\beta_R = \alpha_R R^{\text{true}}$, so that the prior mode matches the true noise variance.

3 Example 1b: Synthetic phase data with nonlinear measurement

Introduction.

3.1 Generative Model

As in Example 1a, the unwrapped phase evolves according to the discrete-time stochastic process

$$\theta_{k+1} = \theta_k + \omega\Delta t + \varepsilon_k, \qquad \varepsilon_k \sim \mathcal{N}(0, 2D_\theta \Delta t), \tag{3.1}$$

for $k = 1, \ldots, N-1$. Now, however, we consider that the phase is measured indirectly through a two-dimensional, nonlinear mapping with additive noise

$$\mathbf{y}_k = \begin{bmatrix} y_{1k} \\ y_{2k} \end{bmatrix} = \begin{bmatrix} \cos\theta_k \\ \sin\theta_k \end{bmatrix} + \boldsymbol{\eta}_k, \qquad \boldsymbol{\eta}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}). \tag{3.2}$$

Equivalently, the measurement model may be written compactly as in terms of the observation function $\mathbf{h}(\theta)$

$$\mathbf{y}_k = \mathbf{h}(\theta_k) + \boldsymbol{\eta}_k, \qquad \mathbf{h}(\theta) = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}. \tag{3.3}$$

Thus, the model represents a drifting–diffusing phase that is embedded in $\mathbb{R}^2$ via its sine and cosine, and corrupted by additive Gaussian noise. All inference proceeds as before, with the only modification being the nonlinear observation map and two-dimensional measurement covariance.

3.2 Inferring the Latent Phase, $p(\theta_{1:N} \mid \mathbf{y}_{1:N}, \lambda)$

In contrast to Example 1a, the observation model is nonlinear. To retain Gaussian filtering and smoothing, we linearize the measurement function about the predicted phase using an extended Kalman filter (EKF). The resulting locally linear–Gaussian model yields an approximate smoothing distribution

$$p(\theta_{1:N} \mid \mathbf{y}_{1:N}, \lambda) \approx \mathcal{N}\Big(\theta_{1:N} \mid \hat{\theta}_{1:N}, \Sigma\Big), \tag{3.4}$$

whose mean and covariance are computed in $\mathcal{O}(N)$ time using a forward EKF pass followed by RTS smoothing.

3.2.1 Extended Kalman Filter (EKF), $p(\theta_k \mid \mathbf{y}_{1:k}, \lambda)$

We approximate the nonlinear measurement model by a first-order Taylor expansion about the one-step-ahead predicted mean $\hat{\theta}_{k|k-1}$ obtained from the forward (extended) Kalman filter. A first-order Taylor expansion about $\hat{\theta}_{k|k-1}$ gives

$$\mathbf{h}(\theta_k) \approx \mathbf{h}(\hat{\theta}_{k|k-1}) + \mathbf{H}_k\big(\theta_k - \hat{\theta}_{k|k-1}\big), \tag{3.5}$$

where the Jacobian of the observation map is

$$\mathbf{H}_k = \left.\frac{\partial \mathbf{h}}{\partial \theta}\right|_{\theta=\hat{\theta}_{k|k-1}} = \begin{bmatrix} -\sin\hat{\theta}_{k|k-1} \\ \cos\hat{\theta}_{k|k-1} \end{bmatrix}. \tag{3.6}$$

The linearized measurement model may therefore be written in affine form as

$$\mathbf{y}_k \approx \mathbf{c}_k + \mathbf{H}_k\theta_k + \boldsymbol{\eta}_k, \qquad \mathbf{c}_k = \mathbf{h}(\hat{\theta}_{k|k-1}) - \mathbf{H}_k\hat{\theta}_{k|k-1}. \tag{3.7}$$

This yields a time-varying linear-Gaussian surrogate model that is consistent with the forward EKF pass. The matrices $\mathbf{H}_k$ (and offsets $\mathbf{c}_k$) computed during that pass are stored and subsequently used for all smoothing and expectation calculations in this example. No further relinearization is performed.

Prediction. Using this linearized model, the predicted phase θ_k conditioned on the prior observations $\mathbf{y}_{1:k-1}$ is normally distributed with mean and variance given by

$$\begin{aligned} \hat{\theta}_{k|k-1} &= \hat{\theta}_{k-1|k-1} + \omega\Delta t \\ P_{k|k-1} &= P_{k-1|k-1} + 2D_\theta\Delta t \end{aligned} \tag{3.8}$$

Update. Given the observation $\mathbf{y}_k$, the updated estimate for the phase θ_k remains Gaussian with mean and variance

$$\begin{aligned} \hat{\theta}_{k|k} &= \hat{\theta}_{k|k-1} + \mathbf{K}_k\big(\mathbf{y}_k - \mathbf{c}_k - \mathbf{H}_k\hat{\theta}_{k|k-1}\big) \\ P_{k|k} &= P_{k|k-1} - \mathbf{K}_k\mathbf{S}_k\mathbf{K}_k^\top \end{aligned} \tag{3.9}$$

where $\mathbf{K}_k = P_{k|k-1}\mathbf{H}_k^\top\mathbf{S}_k^{-1}$ is the Kalman gain, and $\mathbf{S}_k = \mathbf{H}_k P_{k|k-1}\mathbf{H}_k^\top + \mathbf{R}$ is the covariance of the predicted observation.

Initialization. To improve robustness and convergence of the inference procedure, we initialize the phase prior using simple data-driven estimates. First, we obtain an initial estimate of the frequency $\hat{\omega}_{\mathrm{FFT}}$ from the observed time series using a fast Fourier transform (FFT), taking the dominant peak in the power spectrum. Holding this frequency fixed, we then estimate the initial phase by performing a linear regression over a short window of early-time data, approximating the phase evolution as $\theta(t) \approx \theta_1 + \hat{\omega}_{\mathrm{FFT}}(t - t_1)$. This regression yields both a point estimate of the initial phase and an associated uncertainty, computed using a provisional estimate of the measurement noise variance R derived from the prior. The resulting mean and variance define a Gaussian prior on the initial phase, which replaces the nominal prior and is used consistently in all subsequent filtering, smoothing, and parameter inference steps. This initialization strategy leverages the local structure of the data to provide a well-centered and appropriately scaled prior, while remaining computationally inexpensive and independent of the full nonlinear inference procedure.

3.2.2 Backward RTS Smoothing, $p(\theta_k \mid \mathbf{y}_{1:N}, \lambda)$

After the forward EKF pass, the filtered distributions $p(\theta_k \mid \mathbf{y}_{1:k}, \lambda)$ are Gaussian with means $\hat{\theta}_{k|k}$ and variances $P_{k|k}$. A backward Rauch–Tung–Striebel (RTS) pass then yields the smoothed distributions $p(\theta_k \mid \mathbf{y}_{1:N}, \lambda) = \mathcal{N}(\hat{\theta}_{k|N}, P_{k|N})$, where the mean and covariance are given by the recursive update equations (2.18) explained in the previous example.

3.3 Estimating Parameters, $p(\lambda \mid y_{1:N})$

Parameter estimation for the nonlinear measurement model is performed by numerically maximizing the marginal posterior

$$\hat{\lambda} = \arg\max_{\lambda} p(\lambda \mid y_{1:N}), \tag{3.10}$$

where $\lambda = \{\omega, D_\theta, R\}$. The marginal likelihood $p(y_{1:N} \mid \lambda)$ is approximated using the extended Kalman filter (EKF) described above. To estimate the posterior uncertainty, we linearize the observation model using the MAP estimate $\hat{\lambda}$ to create tractable surrogate model that captures the dominant local curvature of the marginal likelihood. This approximate surrogate model can then be treated using the same methods detailed for the previous Example 1a.

3.3.1 Frozen Linearized Model

Let $\hat{\theta}_{k|k-1}$ denote the predicted phase obtained from the EKF at $\hat{\lambda}$. The observation function is linearized about this trajectory,

$$\mathbf{h}(\theta_k) \approx \mathbf{c}_k + \mathbf{H}_k \theta_k, \tag{3.11}$$

where

$$\mathbf{H}_k = \left.\frac{\partial \mathbf{h}}{\partial \theta}\right|_{\hat{\theta}_{k|k-1}}, \qquad \mathbf{c}_k = \mathbf{h}(\hat{\theta}_{k|k-1}) - \mathbf{H}_k \hat{\theta}_{k|k-1}. \tag{3.12}$$

These quantities are computed once at $\hat{\lambda}$ and then held fixed. The resulting surrogate model is

$$\begin{aligned} \theta_{k+1} &= \theta_k + \omega \Delta t + \varepsilon_k, \qquad \varepsilon_k \sim \mathcal{N}(0,\, 2D_\theta \Delta t), \\ \mathbf{y}_k &= \mathbf{c}_k + \mathbf{H}_k \theta_k + \boldsymbol{\eta}_k, \quad \boldsymbol{\eta}_k \sim \mathcal{N}(\mathbf{0},\, \mathbf{R}). \end{aligned} \tag{3.13}$$

3.3.2 MAP Estimation

With $\mathbf{H}_k$ and $\mathbf{c}_k$ fixed, the posterior distribution for λ becomes the posterior of a linear–Gaussian model. The MAP estimate of this surrogate posterior is

$$\hat{\lambda}_{\text{lin}} = \arg\max_{\lambda} p_{\text{lin}}(\lambda \mid y_{1:N}), \tag{3.14}$$

where p_{lin} denotes the posterior under the frozen linearized model. This optimization is inexpensive and is initialized at $\hat{\lambda}$ obtained from the nonlinear EKF objective.

3.3.3 Posterior Uncertainty

Let $\ell_c(\lambda; \theta, y)$ denote the complete–data log likelihood of the frozen linear model. The observed information matrix is obtained using the Louis identity

$$\mathcal{I} = \mathcal{I}_{\text{prior}} + \mathcal{I}_{\text{comp}} - \mathcal{I}_{\text{miss}}, \tag{3.15}$$

where the complete-data information and the missing information are defined as

$$\mathcal{I}_{\text{comp}} = -\,\mathbb{E}\left[\nabla^2_\lambda \ell_c(\lambda; \theta, y) \mid y, \lambda\right], \tag{3.16}$$

$$\mathcal{I}_{\text{miss}} = \text{Cov}[\nabla_\lambda \ell_c(\lambda; \theta, y) \mid y, \lambda]\,. \tag{3.17}$$

These quantities are computed using expectations with respect to the smoothing distribution $p(\theta_{1:N} \mid y_{1:N}, \hat{\lambda}_{\text{lin}})$, which is obtained by applying the Kalman filter and RTS smoother to the frozen linearized model. The Laplace approximation to the posterior covariance of the parameters is then

$$\text{Cov}(\lambda \mid y_{1:N}) \approx \mathcal{I}_{\text{obs}}^{-1}. \tag{3.18}$$

3.3.4 Alternative Approach using Generalized EM

As an alternative to direct numerical maximization of the marginal posterior, we also implemented a generalized expectation–maximization (EM) scheme for parameter estimation. At each iteration, the nonlinear measurement model is first linearized about the current parameter estimate using the extended Kalman filter, and this linearization is held fixed to define a locally consistent linear–Gaussian surrogate model. For this frozen model, the latent phase posterior can be computed exactly using Kalman smoothing, and the expected complete-data log posterior admits closed-form maximization with respect to the parameters $\lambda = \{\omega, D, R\}$. This yields analytic update rules for each parameter analogous to those of Section 2.3.2, which are iterated until convergence. Because the linearization is updated at every iteration, this procedure constitutes a generalized EM algorithm for the original nonlinear model. In practice, we find that this approach converges more rapidly than direct numerical optimization of the EKF-based marginal posterior.

3.4 Quantifying Performance

Linearization condition. The validity of the linear approximation requires that the neglected higher order terms are negligible by comparison

$$\mathbf{h}(\theta_k) = \mathbf{h}(\hat{\theta}_{k|k-1}) + \mathbf{h}'(\hat{\theta}_{k|k-1})(\theta_k - \hat{\theta}_{k|k-1}) + \tfrac{1}{2}\mathbf{h}''(\hat{\theta}_{k|k-1})(\theta_k - \hat{\theta}_{k|k-1})^2 + \dots \tag{3.19}$$

The quadratic terms are negligible when

$$\frac{\|\mathbf{h}''(\hat{\theta}_{k|k-1})\| P_{k|k-1}^{1/2}}{\|\mathbf{h}'(\hat{\theta}_{k|k-1})\|} \ll 1 \tag{3.20}$$

where $\|\cdot\|$ denotes the Euclidean norm, and $P_{k|k-1}^{1/2}$ is the characteristic magnitude of the phase deviation. More generally, the relevant notion of curvature is measured relative to the observation noise. Defining the noise-whitened measurement $\tilde{\mathbf{h}} = \mathbf{R}^{-1/2}\mathbf{h}$, where $\mathbf{R}$ is the measurement covariance, the linearization condition becomes

$$\frac{\|\mathbf{R}^{-1/2}\mathbf{h}''(\hat{\theta}_{k|k-1})\| P_{k|k-1}^{1/2}}{\|\mathbf{R}^{-1/2}\mathbf{h}'(\hat{\theta}_{k|k-1})\|} \ll 1 \tag{3.21}$$

For isotropic noise $\mathbf{R} = R\mathbf{I}_2$, this reduces to the simpler condition above.

For the present harmonic measurement model with unit amplitude and angular frequency ω, we have $\|\mathbf{h}'(\theta_{k|k-1})\| = 1$ and $\|\mathbf{h}''(\theta_{k|k-1})\| = 1$. Assuming isotropic noise R, the linearization condition simplifies to

$$P_{k|k-1}^{1/2} \ll 1 \tag{3.22}$$

Recall from equation (3.8) that the predictive variance $P_{k|k-1}$ is the sum of the filtered variance $P_{k-1|k-1}$ and the process variance $2D_\theta \Delta t$. For small noise ($R \ll D_\theta \Delta t$), the predictive variance is dominated by process noise, so $P_{k|k-1} \approx 2D_\theta \Delta t$, and the linearization condition reduces to $2D_\theta \Delta t \ll 1$. Alternatively, for large noise ($R \gg D_\theta \Delta t$), the predictive variance depends on the measurement uncertainty as $P_{k|k-1} \approx (2D_\theta \Delta t R)^{1/2}$, and the linearization condition reduces to $R \ll 1$. To summarize, we require the following conditions for accurate linearization

$$(2D_\theta \Delta t)^{1/2} \ll 1 \quad \text{and} \quad (2D_\theta \Delta t R)^{1/4} \ll 1 \tag{3.23}$$

Avoiding bias. While uncertainty due to process and measurement noise should be small to ensure accurate linearization, these stochastic contributions should remain large compared to the linearization error to avoid systematic errors in the inferred quantities. Over a single time step, the phase advances by $\theta_k - \hat{\theta}_{k|k-1} = \mathcal{O}(\omega \Delta t)$, so the neglected higher-order terms in the measurement expansion introduce a deterministic error of order $\|\mathbf{h}''(\hat{\theta}_{k|k-1})\|(\omega \Delta t)^2$. When R and $D_\theta \Delta t$ are sufficiently small, this linearization error can exceed the stochastic variability in the measurement and phase dynamics. As a result, the EKF no longer provides a statistically consistent approximation to the true nonlinear model. Instead, the deterministic linearization error is absorbed into the inferred parameters. To avoid these systematic errors, we require that the linearization error be smaller than the measurement noise

$$\|\mathbf{h}''(\hat{\theta}_{k|k-1})\|(\omega \Delta t)^2 \ll \sqrt{R} \tag{3.24}$$

Benchmarking. To assess the impact of violating the linearization conditions, we perform a series of inference experiments in which key dimensionless groups are varied independently while holding all other parameters fixed (Figure S3). When the small-parameter conditions in eqs. (3.23)–(3.24) are satisfied, the MAP estimates for ω, D_θ, and R are accurate and the associated uncertainties, obtained from the Laplace approximation, are well calibrated. In this regime, increasing the number of measurements N reduces the posterior variance without introducing bias, and the estimates converge toward their true values. In contrast, when any of the linearization conditions are violated, systematic deviations emerge that are not captured by the posterior uncertainty.

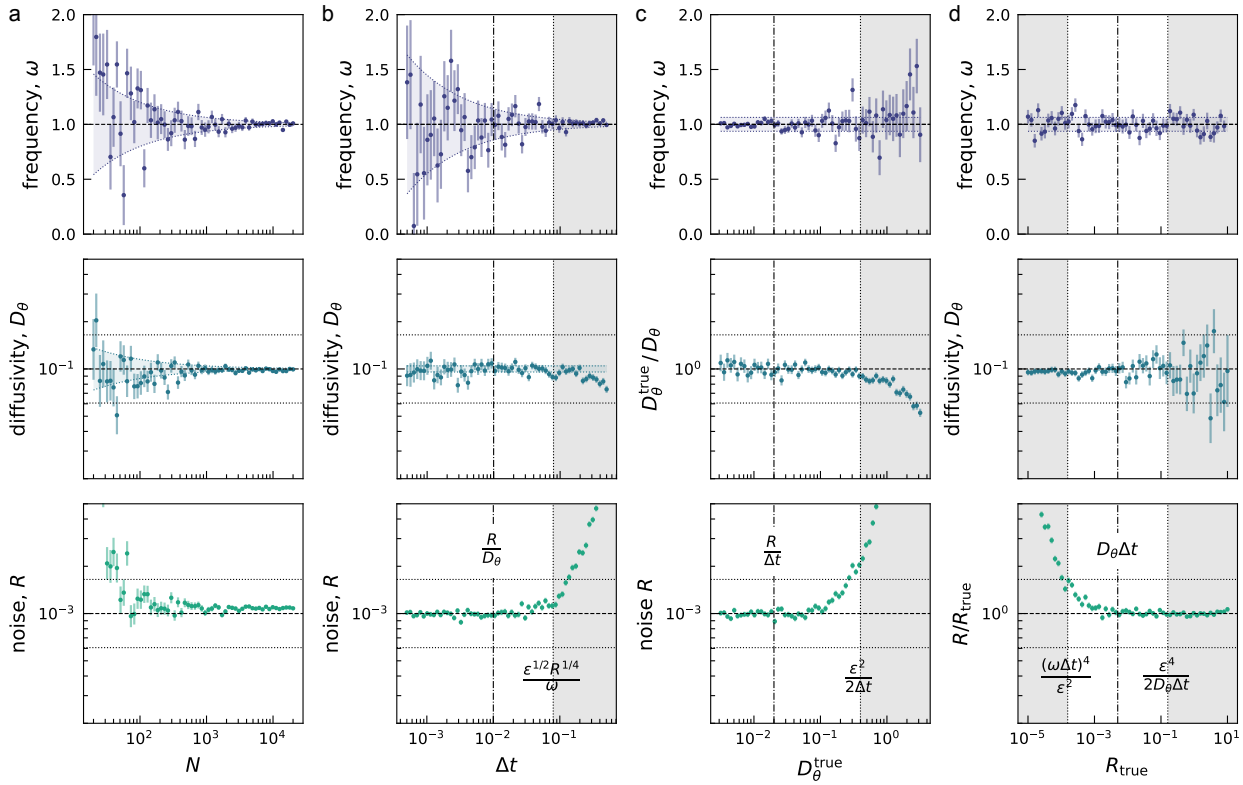


Figure S5: MAP estimates (markers) of the frequency ω, phase diffusivity D_θ, and noise variance R as functions of (a) measurement count N, (b) time step Δt, (c) true phase diffusivity D_θ^{true}, and (d) true noise variance R^{true}. The true values are indicated by horizontal dashed lines, and error bars denote one posterior standard deviation from the Laplace approximation. The true frequency is fixed to $\omega^{\text{true}} = 1$, setting the unit of time to $1/\omega^{\text{true}}$. Shaded regions indicate parameter regimes where the linearization conditions (3.23) and (3.24) break down, defined by $\varepsilon = 0.2 \ll 1$. Unless otherwise specified, parameters are fixed at $N = 1000$, $\Delta t = 0.05$, $D_\theta^{\text{true}} = 0.1$, and $R^{\text{true}} = 0.001$. Panel-specific variations are: (a) varying N; (b) varying Δt; (c) varying D_θ^{true}; and (d) varying R^{true}. Prior hyperparameters are $\mu_\omega = \sigma_\omega = 1$, $\alpha_D = \alpha_R = 2$, $\beta_D = \alpha_D D_\theta^{\text{true}}$, and $\beta_R = \alpha_R R^{\text{true}}$.

4 Example 2: Synthetic state data from Stuart-Landau oscillator

In Examples 1a and 1b, the latent variable was the oscillator phase itself, whose stochastic dynamics was prescribed directly. Here, by contrast, we consider a noisy nonlinear oscillator in which the latent variable is the full dynamical state $x(t) \in \mathbb{R}^d$. The phase is not specified a priori but is instead inferred from the noisy state trajectory and compared with the theoretical predictions of phase reduction.

4.1 Generative Model

We adopt the same generative framework introduced in the main text. The latent state $x(t) \in \mathbb{R}^d$ evolves according to

$$\dot{x} = f(x) + G(x)\,\xi(t), \tag{4.1}$$

where $f(x)$ admits a stable limit cycle $x_0(t)$ of period T, and $\xi(t)$ is Gaussian white noise with

$$\langle \xi(t) \rangle = 0, \qquad \langle \xi(t)\xi(t')^\top \rangle = I\,\delta(t-t'). \tag{4.2}$$

The noise is interpreted in the Stratonovich sense and is assumed to be sufficiently weak that trajectories remain close to the deterministic limit cycle. The observed signal $y(t) \in \mathbb{R}^p$ is obtained from the state through a (generally nonlinear) measurement function

$$y(t) = h(x(t)) + \eta(t), \tag{4.3}$$

where $\eta(t)$ is zero-mean Gaussian measurement noise with covariance

$$\langle \eta(t)\eta(t')^\top \rangle = R\,\delta(t-t'). \tag{4.4}$$

Sampling at times $t_k = t_0 + k\Delta t$ yields

$$y_k = h(x_k) + \eta_k, \qquad \eta_k \sim \mathcal{N}(0, R), \tag{4.5}$$

where $x_k = x(t_k)$ for $k = 1, \ldots, N$.

Stuart-Landau Oscillator. As a concrete realization, we consider the Stuart–Landau oscillator, the normal form of a supercritical Hopf bifurcation. The deterministic dynamics are given by

$$f(x) = \begin{pmatrix} a(1-r^2)x_1 - [1 + b(r^2-1)]x_2 \\ a(1-r^2)x_2 + [1 + b(r^2-1)]x_1 \end{pmatrix}, \tag{4.6}$$

where $x = (x_1, x_2)^\top$ and $r^2 = x_1^2 + x_2^2$. In polar coordinates (r, φ), the dynamics decouple as

$$\dot{r} = a(r - r^3), \tag{4.7}$$

$$\dot{\varphi} = 1 + b(r^2 - 1), \tag{4.8}$$

so that the deterministic system admits a stable limit cycle at $r = 1$ with angular frequency $\omega_0 = 1$.

In this setting, the phase is not directly observed but must be inferred from the noisy state trajectory $x(t)$, providing a direct test of stochastic phase reduction. In this example, we consider the simplest observation model,

$$h(x) = x, \tag{4.9}$$

so that the full state is observed directly up to additive noise.

Stuart–Landau oscillator. We now specialize the general results to the Stuart–Landau model, for which the phase reduction can be analyzed explicitly. In polar coordinates (r, φ), the asymptotic phase is

$$\Theta(x) = \varphi + \frac{b}{a} \ln r. \tag{4.10}$$

The parameter $a > 0$ sets the rate of radial relaxation toward the limit cycle at $r = 1$, while b controls the dependence of angular velocity on amplitude. The ratio b/a therefore quantifies the extent to which transient amplitude fluctuations are converted into phase shifts before they decay.

For isotropic noise with state-space diffusivity D_x, the phase diffusivity can be computed analytically,

$$D_\theta = \left(1 + \frac{b^2}{a^2}\right) D_x, \tag{4.11}$$

where the first term corresponds to direct projection of noise onto the phase direction, and the second arises from amplitude–phase coupling: fluctuations in r are converted into phase fluctuations through the amplitude dependence of the angular velocity. The corresponding noise-induced frequency shift is

$$\Delta\omega = \frac{b}{a} D_x. \tag{4.12}$$

For $b \neq 0$, the isochrons are curved, so that displacements transverse to the limit cycle are not orthogonal to the phase direction. Consequently, amplitude fluctuations are systematically converted into phase fluctuations, leading to both enhanced phase diffusion and a nonzero frequency shift.

4.2 Inference

We can use the exact same inference pipeline used in Example 1b

For the Stuart–Landau oscillator, the transverse dynamics reduce to a single amplitude variable $u(t) = r(t) - 1$, where $r = 1$ is the deterministic limit cycle. For weak noise and small deviations $|u| \ll 1$, the transverse dynamics reduce to the Ornstein–Uhlenbeck process

$$\dot{u} = -2a\, u + \sqrt{2D_x}\, \xi_r(t), \tag{4.13}$$

with relaxation time $(2a)^{-1}$ and stationary variance $\langle u^2 \rangle = D_x/(2a)$. Thus, a controls both the magnitude and correlation time of amplitude fluctuations.

- Case 1: Measurement noise is big $R \gg D_x/2a$
- Case 2: Measurement noise is "amplitude noise" $R = D_x/2a$, in which case we require that $\Delta t \gg (2a)^{-1}$ to allow decorrelation.

5 Example 3: Partial state data from glycolytic oscillator

5.1 Generative model

Noise-free dynamics. As a more nonlinear test case, we consider a two-dimensional glycolytic oscillator (Fig. S6).[9] The noise-free state-space dynamics of $x(t) \in \mathbb{R}^2$ are

$$\dot{x} = f(x) \qquad \Rightarrow \qquad \begin{aligned} \dot{x}_1 &= -x_1 + a x_2 + x_1^2 x_2 \\ \dot{x}_2 &= b - a x_2 - x_1^2 x_2 \end{aligned} \tag{5.1}$$

where x_1 and x_2 denote the dimensionless concentrations of ADP and F6P, respectively, and $a, b > 0$ are kinetic parameters. The system has a fixed point

$$x_1^* = b, \qquad x_2^* = \frac{b}{a + b^2}. \tag{5.2}$$

For the parameter values used below, this fixed point is unstable and the deterministic dynamics approach a stable limit cycle $x_0(t)$ with period T_0 and angular frequency $\omega_0 = 2\pi/T_0$. This deterministic limit cycle defines the noise-free waveform observed in the measurement coordinate.

Noisy dynamics and measurements. We introduce weak stochastic forcing and partial observations using the same generative framework described in the main text. The latent state evolves according to

$$\dot{x} = f(x) + G(x)\,\xi(t), \tag{5.3}$$

where $\xi(t)$ is Gaussian white noise with

$$\langle \xi(t) \rangle = 0, \qquad \langle \xi(t) \xi(t')^\top \rangle = I\,\delta(t - t'). \tag{5.4}$$

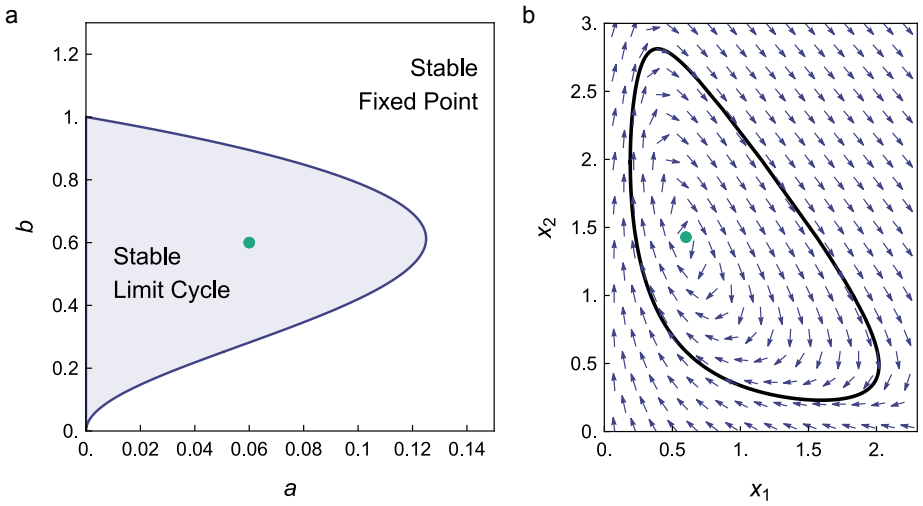


Figure S6: Noise-free dynamics of the glycolytic oscillator. (a) (a, b) parameter space showing the region supporting a stable limit cycle. Simulated data in this section are generated using the parameters $a = 0.06$ and $b = 0.6$ (teal marker). (b) Phase portrait showing the vector field (blue arrows), the unstable fixed point (teak marker), and the stable limit cycle (black curve). The corresponding oscillation frequency is $\omega_0 = 0.6186$.

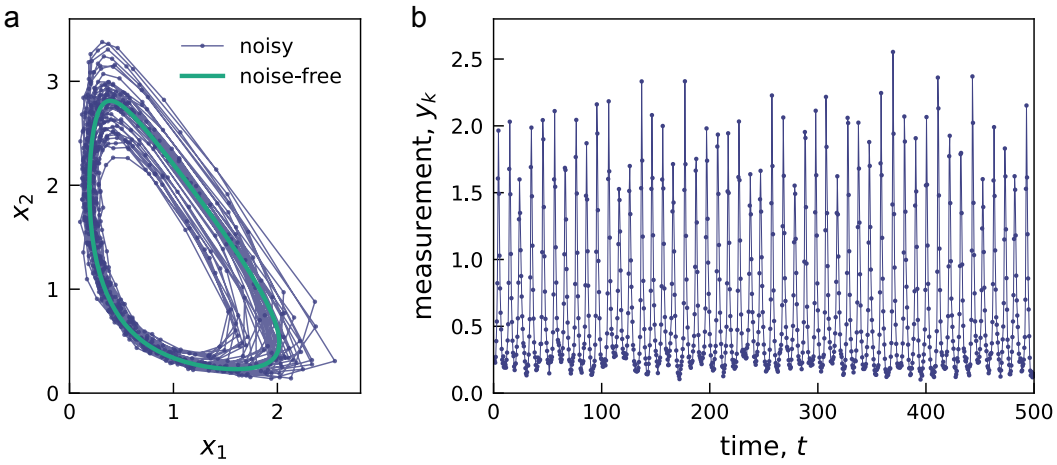


Figure S7: Noisy dynamics of the glycolytic oscillator. (a) Stochastic trajectory in the (x_1, x_2) plane with state-space diffusivity $D_x = 0.001$ and observation interval $\Delta t = 0.5$ (blue markers). The teal curve denotes the noise-free limit cycle. (b) Simulated measurements y_k versus time t_k for $k = 1, \ldots, N$ with $N = 1000$ in the absence of additional measurement noise ($R = 0$).

The noise is interpreted in the Stratonovich sense and is assumed to be sufficiently weak that trajectories remain close to the deterministic limit cycle. In this example, we consider isotropic state-space diffusion with constant diffusivity D_x,

$$G(x) = \sqrt{2D_x}\, I, \tag{5.5}$$

so that the state-space noise covariance is $C = GG^\top = 2D_x I$.

The observed signal $y(t) \in \mathbb{R}^p$ is obtained from the state through the measurement model

$$y(t) = h(x(t)) + \eta(t), \tag{5.6}$$

where $\eta(t)$ is Gaussian measurement noise with covariance

$$\langle \eta(t)\eta(t') \rangle = R\,\delta(t - t'). \tag{5.7}$$

Here, unlike Example 2, only partial state information is observed, and we measure the first state coordinate as

$$h(x) = x_1, \tag{5.8}$$

so that the second state variable remains hidden. Sampling at times $t_k = t_0 + k\Delta t$ yields

$$y_k = h(x_k) + \eta_k, \qquad \eta_k \sim \mathcal{N}(0, R), \tag{5.9}$$

where $x_k = x(t_k)$ for $k = 1, \ldots, N$. An example stochastic trajectory together with the corresponding scalar measurements is shown in Fig. S7.

Reduced phase model. For sufficiently weak state-space noise, the long-time dynamics can be approximated by an effective stochastic phase process

$$\theta_{k+1} = \theta_k + \omega\Delta t + \epsilon_k, \qquad \epsilon_k \sim \mathcal{N}(0, 2D_\theta \Delta t), \tag{5.10}$$

together with a periodic observation model

$$y_k = q(\theta_k) + \eta_k. \tag{5.11}$$

Here $q(\theta)$ denotes the noise-free waveform obtained by projecting the deterministic limit cycle onto the observed coordinate, $q(\theta) = h(x_0(\theta))$. Unlike the previous examples, this waveform is no known a priori and is instead inferred from the data using a truncated Fourier representation,

$$q(\theta) = A_0 + \sum_{m=1}^{M} \left[A_m \cos(m\theta) + B_m \sin(m\theta)\right]. \tag{5.12}$$

Because the phase origin is arbitrary, the representation above is not identifiable without an additional condition: we therefore fix the phase origin by imposing $B_1 = 0$ with the sign convention $B_1 > 0$. This condition anchors $\theta = 0$ to a reference point on the inferred waveform and removes the degeneracy associated with global phase shifts. The unknown parameters are therefore

$$\lambda = \{\omega, D_\theta, R, A_{0:M}, B_{2:M}\}. \tag{5.13}$$

This example extends the previous inference problems in three ways: the phase must be inferred from partial state information, the waveform must be learned directly from the data, and the phase reduction is tested on a more strongly nonlinear oscillator.

5.2 Inferring the Latent Phase, $p(\theta_{1:N} \mid y_{1:N}, \lambda)$

In this example, the measurement model is nonlinear and partially observed. As a result, the phase distribution $p(\theta_k \mid y_{1:k}, \lambda)$ can become multimodal on the phase circle, with multiple phases producing similar observations. In this regime, the linearized measurement model of the extended Kalman filter (Section 3.2.1) can lead the phase distribution to localize about the wrong mode, thereby breaking subsequent linearization approximations and creating systematic errors in the inferred phase trajectory. These errors are particularly pronounced when the waveform $q(\theta)$ is strongly non-sinusoidal, since local ambiguities in the observation model can persist over extended portions of the oscillation cycle. To address this issue, we infer the latent phase using a global phase filter defined directly on the circle.

5.2.1 Global Phase Filter

Rather than approximating the posterior by a single Gaussian distribution, we discretize the phase interval into L uniformly spaced grid points

$$\theta^{(j)} = \frac{2\pi j}{N_\theta}, \qquad j = 0, \ldots, N_\theta - 1, \tag{5.14}$$

and represent the posterior by a probability vector $p_k^{(j)} \approx p(\theta_k = \theta^{(j)} \mid y_{1:k})$. For notational simplicity, we omit explicit conditioning on the parameters λ throughout this section. The latent phase dynamics are modeled by the diffusion process of equation (5.10), which defines a wrapped Gaussian transition kernel

$$T_{ij} = p(\theta_k = \theta^{(j)} \mid \theta_{k-1} = \theta^{(i)}) \propto \sum_{n=-\infty}^{\infty} \exp\left[-\frac{\left(\theta^{(j)} - \theta^{(i)} - \omega\Delta t + 2\pi n\right)^2}{4D_\theta \Delta t}\right]. \tag{5.15}$$

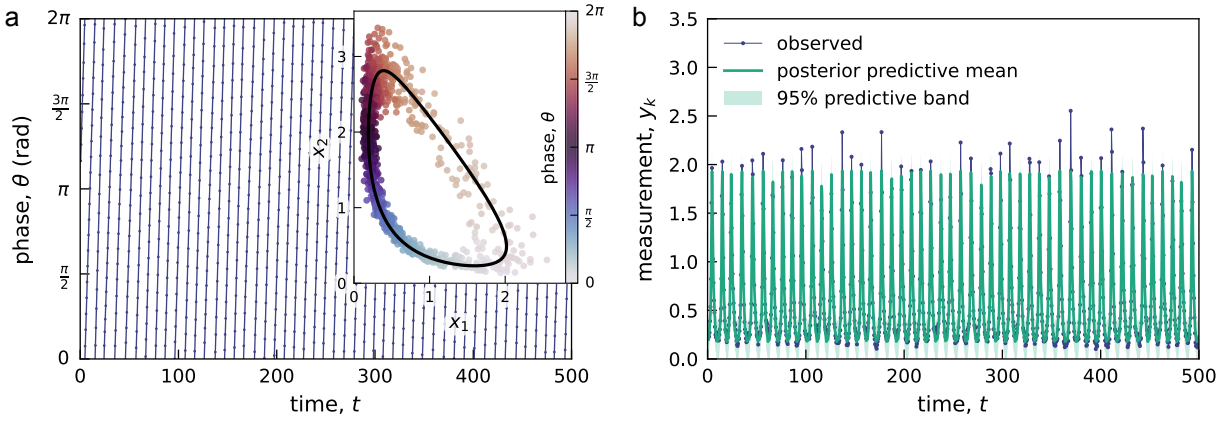


Figure S8: Global phase filter. (a) Phase θ inferred from the measurement data in Fig. S7 using the MAP parameter estimates $\hat{\omega} = 0.6174$ and $\hat{D}_\theta = 0.00577$. The effective noise magnitude used to regularize inference is $R_{\text{eff}} = 0.01$. The inset shows the full oscillator state (x_1, x_2) colored by the inferred phase. (b) Posterior predictive distribution for the measurements y_k obtained from the inferred phase dynamics and waveform model. The teal curve denotes the posterior predictive mean, while the shaded region indicates the 95% predictive interval. Observed measurements are shown as blue markers.

For sufficiently small sampling interval Δt, such that typical phase increments satisfy $\sqrt{2D_\theta \Delta t} \ll 2\pi$, the dominant contribution arises from the winding number corresponding to the shortest angular displacement. In this limit, the transition probability is well approximated by

$$T_{ij} \propto \exp\left(-\frac{\Delta\theta_{ij}^2}{4D_\theta \Delta t}\right) \quad \text{with} \quad \Delta\theta_{ij} = \text{wrap}\Big(\theta^{(j)} - \theta^{(i)} - \omega\Delta t\Big)\,, \tag{5.16}$$

and $\text{wrap}(\phi)$ denotes the unique representative of ϕ in the interval $[-\pi, \pi)$. In practice, the transition matrix is normalized row-wise so that $\sum_j T_{ij} = 1$.

The forward filtering recursion consists of prediction and measurement update steps analogous to the standard hidden Markov model filter. Given the filtered distribution $p(\theta_{k-1} \mid y_{1:k-1})$ at time step $k-1$, the predictive distribution is

$$p(\theta_k = \theta^{(j)} \mid y_{1:k-1}) = \sum_i T_{ij}\, p(\theta_{k-1} = \theta^{(i)} \mid y_{1:k-1}). \tag{5.17}$$

Updating this distribution on the measurement y_k, we obtain the posterior

$$p(\theta_k = \theta^{(j)} \mid y_{1:k}) \propto p(y_k \mid \theta_k = \theta^{(j)})\, p(\theta_k = \theta^{(j)} \mid y_{1:k-1}) \tag{5.18}$$

where the Gaussian measurement likelihood is

$$p(y_k \mid \theta_k = \theta^{(j)}) = \frac{1}{\sqrt{2\pi R}} \exp\left[-\frac{1}{2R}\left(y_k - q(\theta^{(j)})\right)^2\right]. \tag{5.19}$$

In practice, the predictive and posterior distributions are normalized numerically at each time step. The corresponding normalization constants provide the conditional likelihood factors $p(y_k \mid y_{1:k-1}, \lambda)$, whose product yields the marginal likelihood of the observations (see below).

After the forward pass, smoothed posterior distributions are obtained using a backward recursion analogous to the Rauch–Tung–Striebel smoother. The recursive relation (2.14) can be rewritten in an equivalent formulation as

$$p(\theta_k \mid y_{1:N}) = p(\theta_k \mid y_{1:k})\beta_k(\theta_k), \tag{5.20}$$

where $\beta_k(\theta_k) = p(y_{k+1:N} \mid \theta_k)$ is the backward message distribution commonly used in hidden Markov models. This distribution satisfies the recursive relationship

$$\beta_k(\theta_k) = \int p(\theta_{k+1} \mid \theta_k)p(\theta_{k+1} \mid \theta_{k+1})\beta_{k+1}(\theta_{k+1})\, d\theta_{k+1} \tag{5.21}$$

with terminal condition $\beta_N(\theta_N) = 1$. In practice, the backward recursion is evaluated numerically on the discrete phase grid and normalized at each time step for numerical stability. The resulting posterior distributions remain globally defined on the phase circle and naturally accommodate multimodal uncertainty arising from partial observations and nonlinear waveforms. In contrast to local Gaussian filters, the global phase filter preserves competing phase hypotheses simultaneously and therefore avoids spurious phase slips associated with local linearization errors.

5.2.2 Marginal Likelihood, $p(y_{1:N} \mid \lambda)$

The global phase filter also provides a natural approximation of the marginal likelihood $p(y_{1:N} \mid \lambda)$, which is required for Bayesian parameter inference. Using the chain rule,

$$p(y_{1:N} \mid \lambda) = \prod_{k=1}^{N} p(y_k \mid y_{1:k-1}, \lambda), \tag{5.22}$$

where each factor corresponds to the normalization constant appearing in the filtering recursion. Specifically, the predictive distribution $p(\theta_k \mid y_{1:k-1}, \lambda)$ defines the conditional likelihood

$$p(y_k \mid y_{1:k-1}, \lambda) = \int p(y_k \mid \theta_k, \lambda)\, p(\theta_k \mid y_{1:k-1}, \lambda)\, d\theta_k. \tag{5.23}$$

On the discrete phase grid, this integral becomes

$$p(y_k \mid y_{1:k-1}, \lambda) \approx \sum_j p(y_k \mid \theta_k = \theta^{(j)}, \lambda)\, p(\theta_k = \theta^{(j)} \mid y_{1:k-1}, \lambda). \tag{5.24}$$

The filtering update (5.18) can therefore be written as

$$p(\theta_k = \theta^{(j)} \mid y_{1:k}, \lambda) = \frac{p(y_k \mid \theta_k = \theta^{(j)}, \lambda)\, p(\theta_k = \theta^{(j)} \mid y_{1:k-1}, \lambda)}{p(y_k \mid y_{1:k-1}, \lambda)}, \tag{5.25}$$

where the denominator is precisely the normalization constant required to ensure that the posterior distribution sums to unity. Consequently, the log marginal likelihood is obtained directly from the filtering recursion as

$$\log p(y_{1:N} \mid \lambda) = \sum_{k=1}^{N} \log p(y_k \mid y_{1:k-1}, \lambda). \tag{5.26}$$

This quantity provides a globally consistent objective function for Bayesian parameter inference even when the posterior phase distribution is strongly non-Gaussian or multimodal.

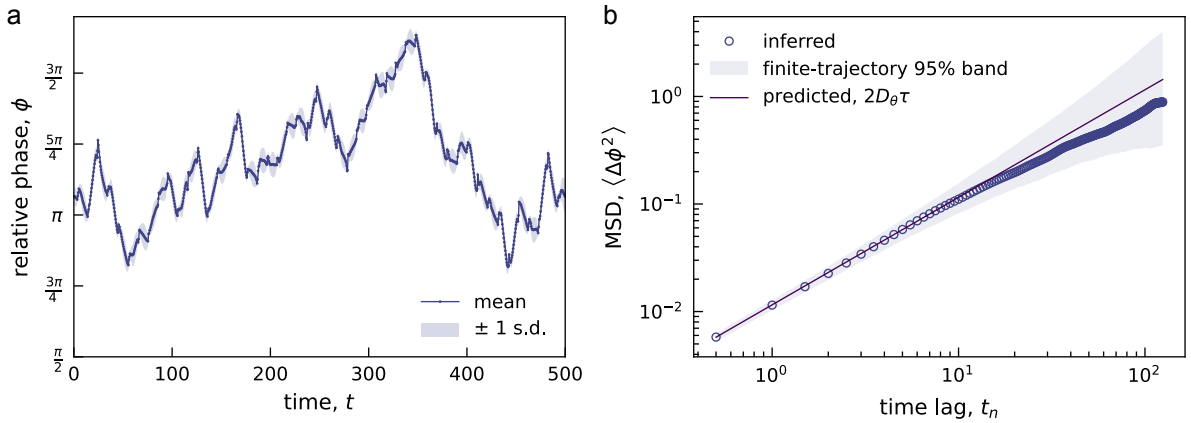


Figure S9: (a) Relative phase $\phi = \theta - \omega t$ vinferred by the global phase filter from the data in Fig. XXX as a function of time. (b) Mean squared displacement (MSD) of the relative phase.

5.2.3 Sampling Phase Trajectories

We can sample phase trajectories from the smoothing distribution $p(\theta_{1:N} \mid y_{1:N}, \lambda)$ using the Forward-Filter Backward-Sampling (FFBS) algorithm. At the final time step, we sample

$$\theta_N \sim p(\theta_N \mid y_{1:N}, \lambda), \tag{5.27}$$

which is approximated by the discrete distribution $\alpha_k(i) = p(\theta_k = \theta^{(i)} \mid y_{1:k}, \lambda)$ for $k = N$. We then proceed backward for $k = N-1, \ldots, 1$. Conditioned on the sampled state θ_{k+1}, the backward conditional distribution is

$$p(\theta_k \mid \theta_{k+1}, y_{1:N}, \lambda) = p(\theta_k \mid \theta_{k+1}, y_{1:k}, \lambda), \tag{5.28}$$

since conditioning on θ_{k+1} removes dependence on future observations $y_{k+2:N}$. Using Bayes' theorem and the Markov property,

$$p(\theta_k \mid \theta_{k+1}, y_{1:N}, \lambda) \propto p(\theta_{k+1} \mid \theta_k, \lambda) p(\theta_k \mid y_{1:k}, \lambda). \tag{5.29}$$

On the phase grid, this conditional distribution becomes

$$p(\theta_k = \theta^{(i)} \mid \theta_{k+1} = \theta^{(j)}, y_{1:N}, \lambda) \propto T_{ij}\, \alpha_k(i), \tag{5.30}$$

where T_{ij} is the discrete transition matrix associated with the wrapped diffusion kernel. After normalization over i, we sample θ_k from this discrete conditional distribution. Iterating this recursive sampling procedure, the resulting trajectories provide approximate draws from the full posterior smoothing distribution. These samples can be used to estimate expectations of nonlinear dynamical observables.

5.2.4 Mean Squared Displacement

Under the smoothing distribution $p(\theta_{1:N} \mid y_{1:N}, \lambda)$, the drift-subtracted mean squared displacement (MSD) of the latent phase trajectory for a lag time $t_n = n\Delta t$ is defined as

$$\mathrm{MSD}(n) = \frac{1}{N-n} \sum_{k=1}^{N-n} \mathbb{E}\Big[\mathrm{wrap}(\theta_{k+n} - \theta_k - \omega t_n)^2 \mid y_{1:N}, \lambda\Big], \tag{5.31}$$

where wrap$(\cdot)$ maps phase differences onto the interval $(-\pi, \pi]$. For the global phase filter, the smoothing distribution is represented on a discrete phase grid and does not admit a simple closed-form covariance representation analogous to the Kalman smoother. Instead, we estimate the MSD using sampled trajectories

$$\theta_{1:N}^{(s)} \sim p(\theta_{1:N} \mid y_{1:N}, \lambda), \tag{5.32}$$

generated using the Forward-Filter Backward-Sampling (FFBS) procedure described above. For each sampled trajectory and lag n, we compute the drift-subtracted phase increment

$$\Delta\theta_k^{(s)}(n) = \mathrm{wrap}\Big(\theta_{k+n}^{(s)} - \theta_k^{(s)} - \omega t_n\Big). \tag{5.33}$$

The corresponding Monte Carlo estimator is

$$\mathrm{MSD}(n) \approx \frac{1}{S(N-n)} \sum_{s=1}^{S} \sum_{k=1}^{N-n} \Big[\Delta\theta_k^{(s)}(n)\Big]^2, \tag{5.34}$$

where S denotes the number of sampled trajectories. For pure phase diffusion, the theoretical drift-subtracted MSD is $\mathrm{MSD}_0(t) = 2D_\theta t$. Figure S9b shows that the inferred MSD obtained from the sampled smoothing trajectories is in good agreement with this theoretical expectation over the range of lag times considered.

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5.3 Estimating Parameters, $p(\lambda \mid y_{1:N})$

5.3.1 Prior Distribution, $p(\lambda)$

To regularize parameter estimation, we place prior distributions on the unknown parameters of the reduced phase model. In Example 3, the parameter vector includes both the stochastic phase-dynamics parameters and the Fourier coefficients defining the learned waveform,

$$\lambda = \{\omega, D_\theta, R, A_0, \ldots, A_M, B_2, \ldots, B_M\}. \tag{5.35}$$

The coefficient B_1 is not inferred, as the constraint $B_1 = 0$ fixes the arbitrary phase origin of the waveform. As in the previous examples, we assign a Gaussian prior to the frequency and inverse-gamma priors to the phase diffusivity and measurement-noise variance,

$$\omega \sim \mathcal{N}(\mu_\omega, \sigma_\omega^2), \qquad D_\theta \sim \text{Inv-Gamma}(\alpha_D, \beta_D), \qquad R \sim \text{Inv-Gamma}(\alpha_R, \beta_R). \tag{5.36}$$

The inverse-gamma hyperparameters may be chosen to specify the prior mean $\beta/(\alpha-1)$ and variance $\beta^2/[(\alpha-1)^2(\alpha-2)]$, when these moments exist. For the waveform, we assign a separate Gaussian prior to the mean level A_0,

$$A_0 \sim \mathcal{N}(\mu_A, \sigma_A^2), \tag{5.37}$$

and use a common zero-mean Gaussian prior for all nonconstant Fourier coefficients,

$$\begin{aligned} A_m &\sim \mathcal{N}(0, \sigma_q^2), \qquad m = 1, \ldots, M, \\ B_m &\sim \mathcal{N}(0, \sigma_q^2), \qquad m = 2, \ldots, M. \end{aligned} \tag{5.38}$$

Here σ_q is chosen to be comparable to the characteristic amplitude of the measured oscillations. This weakly regularizes the inferred waveform while allowing the data to determine its shape.

Thus the full prior factorizes as

$$p(\lambda) = p(\omega)\, p(D_\theta)\, p(R)\, p(A_0) \prod_{m=1}^{M} p(A_m) \prod_{m=2}^{M} p(B_m). \tag{5.39}$$

In log coordinates, $\lambda = \{\omega, \log D_\theta, \log R, A_0, \ldots, A_M, B_2, \ldots, B_M\}$, the posterior density includes the corresponding Jacobian factors for D_θ and R. Up to an additive constant, the negative log prior is

$$\begin{aligned} -\log p(\lambda) = \frac{1}{2}\left(\frac{\omega - \mu_\omega}{\sigma_\omega}\right)^2 + \alpha_D \log D_\theta + \frac{\beta_D}{D_\theta} + \alpha_R \log R + \frac{\beta_R}{R} \\ + \frac{1}{2}\left(\frac{A_0 - \mu_A}{\sigma_A}\right)^2 + \frac{1}{2\sigma_q^2}\sum_{m=1}^{M} A_m^2 + \frac{1}{2\sigma_q^2}\sum_{m=2}^{M} B_m^2. \end{aligned} \tag{5.40}$$

5.3.2 MAP Estimation, $\hat{\lambda}$

We consider three complementary strategies for computing the maximum a posteriori (MAP) estimate $\hat{\lambda}$. First, gradient-free optimization directly maximizes the log posterior using only function evaluations and is simple and robust in low dimensions. Second, gradient-based optimization leverages the analytic posterior gradient obtained via the Louis identity, enabling efficient quasi-Newton updates with rapid convergence near the optimum. Third, the expectation–maximization (EM) algorithm alternates between smoothing and closed-form parameter updates under conjugate priors, yielding monotonic ascent of the posterior. In practice, supplying the analytic gradient to a quasi-Newton method (e.g., BFGS) provides the best computational performance, while EM offers interpretability and guaranteed ascent but typically converges more slowly.

Gradient-free optimization. Maximum a posteriori (MAP) parameter estimation is performed by maximizing the log posterior

$$\mathcal{L}(\lambda) = \log p(\lambda \mid y_{1:N}) = \log p(y_{1:N} \mid \lambda) + \log p(\lambda) + \text{constant}, \tag{5.41}$$

where $p(y_{1:N} \mid \lambda)$ is the marginal likelihood (5.26) obtained from the global phase filter, and $p(\lambda)$ is the prior distribution (5.40) described above. A straightforward approach to MAP estimation is to maximize $\mathcal{L}(\lambda)$ using gradient-free optimization methods. In practice, we optimize over the transformed parameters

$$\lambda = \{\omega, \log D_\theta, \log R, A_0, \ldots, A_M, B_2, \ldots, B_M\}, \tag{5.42}$$

which enforces positivity of D_θ and R while improving numerical conditioning. The posterior distribution is therefore parameterized directly in terms of the transformed variables, including the corresponding Jacobian factors for D_θ and R discussed above. Among the derivative-free methods available in `scipy.optimize`, we find Powell's method to converge reliably and efficiently for this problem.

In practice, we often fix the measurement-noise covariance to a prescribed effective value, $R = R_{\text{eff}}$, rather than estimating it simultaneously with the phase diffusivity. This provides an important regularization strategy when the reduced phase model is only approximate. For sufficiently weak state-space noise, the observations remain localized near the deterministic waveform and the residual fluctuations are well described by independent measurement noise. At larger state-space diffusivities, however, the observed trajectories deviate substantially from the noise-free limit cycle due to amplitude fluctuations and other transverse dynamics neglected by the phase reduction. These fluctuations are generally correlated in time and therefore violate the assumptions of the reduced observation model. If R is inferred directly under these conditions, the optimization tends to attribute short-time discrepancies between the observations and the waveform to phase fluctuations, leading to inflated estimates of D_θ. Fixing R_{eff} instead introduces a controlled tolerance between the data and the predicted waveform: rapidly varying deviations are absorbed into the effective observation noise, while the inferred phase dynamics are determined primarily by the slower fluctuations that accumulate over many cycles and more accurately reflect phase diffusion.

Expectation Maximization (EM) algorithm. As an alternative to direct numerical optimization of the posterior $\mathcal{L}(\lambda)$, we also implemented an expectation–maximization (EM) algorithm for joint inference of the phase dynamics and waveform parameters. In the **E-step**, we compute the smoothing distribution $p(\theta_{1:N} \mid y_{1:N}, \lambda^{(m)})$ using the global phase filter and backward recursion described above. From this distribution, we evaluate the expected sufficient statistics associated with the phase increments and waveform residuals. In the subsequent **M-step**, we maximize the expected log posterior

$$Q(\lambda \mid \lambda^{(m)}) = \mathbb{E}\Big[\mathcal{L}(\lambda) \mid y_{1:N}, \lambda^{(m)}\Big], \tag{5.43}$$

with respect to the parameters

$$\lambda = \{\omega, D_\theta, R, A_0, \ldots, A_M, B_2, \ldots, B_M\}. \tag{5.44}$$

Conditioned on the latent phase trajectory, the waveform model

$$y_k = q(\theta_k) + \eta_k \tag{5.45}$$

is linear in the Fourier coefficients. As a result, the updates for A_m and B_m reduce to a regularized weighted least-squares problem, while the updates for ω, D_θ, and R retain the same form as in the previous examples. The phase-origin constraint $B_1 = 0$ is imposed throughout the optimization.

When the effective observation noise is fixed, $R = R_{\text{eff}}$, the M-step updates only the phase-dynamics and waveform parameters. In practice, we find that the EM algorithm converges reliably and substantially faster than direct optimization of the marginal posterior, particularly when simultaneously inferring high-dimensional waveform representations with many Fourier modes.

5.3.3 Posterior Uncertainty via Laplace Approximation, $\mathcal{I}(\hat{\lambda})$

Uncertainty in the MAP estimate $\hat{\lambda}$ is quantified using a Laplace approximation to the posterior. Expanding the log posterior $\mathcal{L}(\lambda)$ to second order about $\hat{\lambda}$ yields a local Gaussian approximation

$$p(\lambda \mid y) \approx \mathcal{N}\Big(\hat{\lambda}, \mathcal{I}^{-1}(\hat{\lambda})\Big) \tag{5.46}$$

where the posterior information matrix is the negative Hessian of the log posterior evaluated at the optimum,

$$\mathcal{I}(\hat{\lambda}) = -\nabla^2_\lambda \mathcal{L}(\lambda)\big|_{\lambda=\hat{\lambda}}. \tag{5.47}$$

The diagonal entries of $\mathcal{I}^{-1}$ provide approximate posterior variances, while the full matrix captures parameter correlations. We consider both numerical and analytical strategies for evaluating $\mathcal{I}(\hat{\lambda})$.

Numerical Hessian. The most direct approach is to compute the Hessian of $\mathcal{L}(\lambda)$ by finite differences of either the log posterior itself or its gradient. In practice, central-difference schemes applied to the analytic score $\nabla_\lambda \mathcal{L}(\lambda)$ provide improved stability and second-order accuracy. This approach is straightforward to implement and makes minimal use of model structure, but it requires multiple evaluations of the Kalman smoother and can become computationally expensive as the parameter dimension grows.

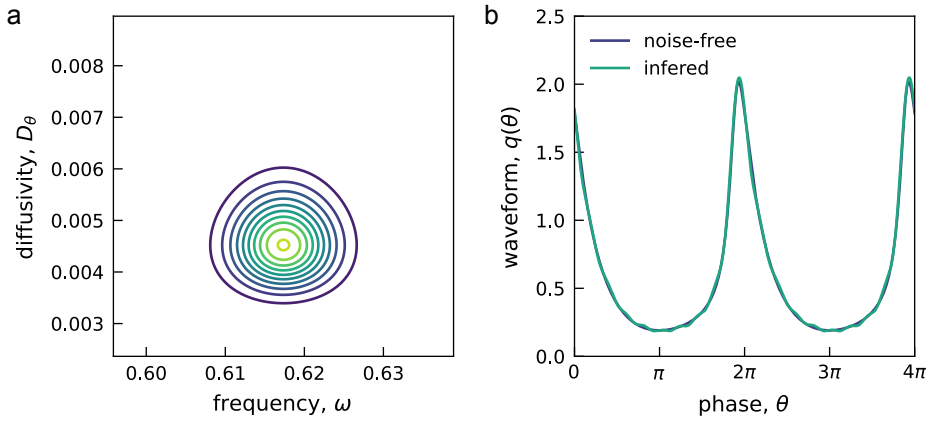


Figure S10: Caption

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